Chapter IV

THE TEST PROBLEMS

Using the documentation scheme as proposed in Section 4 of Chapter I, we present a detailed description of 114 problems for testing nonlinear programming algorithms. The corresponding FORTRAN subroutines are available on request. Extensive lists of constant data are shown in Appendix A.

Originally, a set of 126 test problems has been implemented. But we could not solve 12 problems with the desired degree of accuracy and check the optimality conditions at the computed solution. In particular, we did not succeed in solving some big problems with more than 20 variables. The underlying idea for this collection of test problems is to present only those examples which could be implemented and solved successfully in the sense that the Kuhn-Tucker and second order optimality conditions are verified numerically.

In many test problems given in the literature, we find lower bounds of the kind $x_i \geq 0$ together with a division by x_i in one of the problem functions. In most of these cases, the zero bounds are replaced by some small positive numbers without influencing the optimal solution. As a consequence, many irregular problems are transformed into regular ones in the corresponding subroutines.

For investigating the active sets $I(x^*)$, the reader should pay attention to the succession of the constraints which is defined in the following form, cf. (2): Inequalities, lower bounds, upper bounds, equalities. In case of several local optimal solutions x^*, the remaining data $f(x^*)$, $r(x^*)$, etc., are only presented for the first one. If real numbers are given with one or two digits only, it is assumed that this is the precise value, i.e. that all other digits are zero. The optimal solutions are not always the same as those found in the literature. We suppose that we got either another local solution or, more probably, that there are printing errors in the source.

W. Hock et al., *Test Examples for Nonlinear Programming Codes*
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PROBLEM:	1
CLASSIFICATION:	PBR-T1-1
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 1$

OBJECTIVE FUNCTION:

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

CONSTRAINTS:

$$-1.5 \leq x_2$$

START:	x_o	=	(-2 , 1)	(feasible)
	$f(x_o)$	=	909	
SOLUTION:	x^*	=	(1 , 1)	
	$f(x^*)$	=	0	
	$r(x^*)$	=	0	
	$e(x^*)$	=	-	
	μ	=	0	
	$I(x^*)$	=	-	
	u^*_{max}/u^*_{min}	=	-	
	$\lambda^*_{max}/\lambda^*_{min}$	=	.25E 4	

PROBLEM:	2
CLASSIFICATION:	PBR-T1-2
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 1$

OBJECTIVE FUNCTION:

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

CONSTRAINTS:

$$1.5 \leq x_2$$

START:	x_o	$= (-2 , 1)$	(not feasible)
	$f(x_o)$	$= 909$	

SOLUTION:	x^*	$= (2a \cos(\frac{1}{3} \arccos\frac{1}{b}) , 1.5)$	
	$f(x^*)$	$= .05042\ 61879$	$a = (598/1200)^{1/2}$
	$r(x^*)$	$= 0$	$b = 400\ a^3$
	$e(x^*)$	$= .13E-7$	
	μ	$= 1$	
	$I(x^*)$	$= (1)$	
	u^*_{max}/u^*_{min}	$= .1833/.1833 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 200/200 = 1$	

PROBLEM:	3
CLASSIFICATION:	QBR-T1-1
SOURCE:	Schuldt [56]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 1$

OBJECTIVE FUNCTION:

$$f(x) = x_2 + 10^{-5}(x_2 - x_1)^2$$

CONSTRAINTS:

$$0 \leq x_2$$

START:	x_o	=	(10 , 1)	(feasible)
	$f(x_o)$	=	1.00081	

SOLUTION:	x^*	=	(0 , 0)
	$f(x^*)$	=	0
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	1
	$I(x^*)$	=	(1)
	u^*_{max}/u^*_{min}	=	1.0000/1.0000 = 1
	$\lambda^*_{max}/\lambda^*_{min}$	=	.20E-4/.20E-4 = 1

PROBLEM:	4
CLASSIFICATION:	PBR-T1-3
SOURCE:	Asaadi [1]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 2$

OBJECTIVE FUNCTION:

$$f(x) = \frac{1}{3}(x_1 + 1)^3 + x_2$$

CONSTRAINTS:

$$1 \le x_1$$

$$0 \le x_2$$

START:	x_o	=	(1.125 , .125)	(feasible)
	$f(x_o)$	=	3.323568	
SOLUTION:	x^*	=	(1 , 0)	
	$f(x^*)$	=	8/3	
	$r(x^*)$	=	0	
	$e(x^*)$	=	0	
	μ	=	2	
	$I(x^*)$	=	(1 , 2)	
	u^*_{max}/u^*_{min}	=	4.0000/1.0000 = 4.00	
	$\lambda^*_{max}/\lambda^*_{min}$	=	-	

PROBLEM:	5
CLASSIFICATION:	GBR-T1-1
SOURCE:	McCormick [41]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = \sin(x_1 + x_2) + (x_1 - x_2)^2 - 1.5x_1 + 2.5x_2 + 1$$

CONSTRAINTS:

$$-1.5 \le x_1 \le 4$$

$$-3 \le x_2 \le 3$$

START:	x_o	=	$(0 , 0)$	(feasible)
	$f(x_o)$	=	1	
SOLUTION:	x^*	=	$(-\frac{\pi}{3} + \frac{1}{2} , -\frac{\pi}{3} - \frac{1}{2})$	
	$f(x^*)$	=	$-\frac{1}{2}\sqrt{3} - \frac{\pi}{3}$	
	$r(x^*)$	=	0	
	$e(x^*)$	=	-	
	μ	=	0	
	$I(x^*)$	=	-	
	u^*_{max}/u^*_{min}	=	-	
	$\lambda^*_{max}/\lambda^*_{min}$	=	$4.00/1.73 = 2.31$	

PROBLEM:	6
CLASSIFICATION:	QQR-T1-1
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (1 - x_1)^2$$

CONSTRAINTS:

$$10(x_2 - x_1^2) = 0$$

START:	x_o	=	(-1.2 , 1)	(not feasible).
	$f(x_o)$	=	4.84	
SOLUTION:	x^*	=	(1 , 1)	
	$f(x^*)$	=	0	
	$r(x^*)$	=	0	
	$e(x^*)$	=	0	
	μ	=	0	
	$I(x^*)$	=	-	
	u^*_{max}/u^*_{min}	=	0	
	$\lambda^*_{max}/\lambda^*_{min}$	=	.40/.40 = 1	

PROBLEM:	7
CLASSIFICATION:	GPR-T1-1
SOURCE:	Miele e.al. [44,45]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = \ln(1 + x_1^2) - x_2$$

CONSTRAINTS:

$$(1 + x_1^2)^2 + x_2^2 - 4 = 0$$

START:	x_o	$= (2 , 2)$	(not feasible)
	$f(x_o)$	$= \ln 5 - 2$	
SOLUTION:	x^*	$= (0 , \sqrt{3})$	
	$f(x^*)$	$= -\sqrt{3}$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= .21E-24$	
	μ	$= 0$	
	$I(x^*)$	$= -$	
	u^*_{max}/u^*_{min}	$= .2887/.2887 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 3.15/3.15 = 1$	

PROBLEM:	8
CLASSIFICATION:	CQR-T1-1
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = -1$$

CONSTRAINTS:

$$x_1^2 + x_2^2 - 25 = 0$$

$$x_1 x_2 - 9 = 0$$

START:	x_o	$= (2 , 1)$	(not feasible)
	$f(x_o)$	$= -1$	

SOLUTION:	x^*	$= (a , \frac{9}{a}) , (-a , -\frac{9}{a}) , (b , \frac{9}{b}) , (-b , -\frac{9}{b})$	
	$f(x^*)$	$= -1$	
	$r(x^*)$	$= 0$	$a = \sqrt{\frac{25 + \sqrt{301}}{2}}$
	$e(x^*)$	$= 0$	
	μ	$= 0$	$b = \sqrt{\frac{25 - \sqrt{301}}{2}}$
	$I(x^*)$	$= -$	
	u^*_{max}/u^*_{min}	$= 0$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= -$	

PROBLEM:	9
CLASSIFICATION:	GLR-T1-1
SOURCE:	Miele e.al. [44]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1(1)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = \sin(\pi x_1/12)\cos(\pi x_2/16)$$

CONSTRAINTS:

$$4x_1 - 3x_2 = 0$$

START:	x_o	=	(0 , 0)	(feasible)
	$f(x_o)$	=	0	
SOLUTION:	x^*	=	$(12k - 3 , 16k - 4)$, $k=0, \pm 1, \pm 2, \ldots$	
	$f(x^*)$	=	$-.5$	
	$r(x^*)$	=	0	
	$e(x^*)$	=	.73E-12	
	μ	=	0	
	$I(x^*)$	=	-	
	u^*_{max}/u^*_{min}	=	.03272/.03272 = 1	
	$\lambda^*_{max}/\lambda^*_{min}$	=	.049/.049 = 1	

PROBLEM:	10
CLASSIFICATION:	LQR-T1-1
SOURCE:	Biggs [10]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = x_1 - x_2$$

CONSTRAINTS:

$$-3x_1^2 + 2x_1x_2 - x_2^2 + 1 \geq 0$$

START:	x_o	= (-10 , 10)	(not feasible)
	$f(x_o)$	= -20	
SOLUTION:	x^*	= (0 , 1)	
	$f(x^*)$	= -1	
	$r(x^*)$	= 0	
	$e(x^*)$	= .92E-11	
	μ	= 1	
	$I(x^*)$	= (1)	
	u^*_{max}/u^*_{min}	= .5000/.5000 = 1	
	$\lambda^*_{max}/\lambda^*_{min}$	= 1.00/1.00 = 1	

PROBLEM:	11
CLASSIFICATION:	QQR-T1-2
SOURCE:	Biggs [10]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 5)^2 + x_2^2 - 25$$

CONSTRAINTS:

$$-x_1^2 + x_2 \geq 0$$

START:	x_o	$= (4.9 , .1)$.	(not feasible)
	$f(x_o)$	$= -24.98$	

SOLUTION:	x^*	$= ((a - \frac{1}{a})/\sqrt{6} , (a^2 - 2 + a^{-2})/6)$	
	$f(x^*)$	$= -8.49846\ 4223$	
	$r(x^*)$	$= 0$	$a = 7.5\sqrt{6} + \sqrt{338.5}$
	$e(x^*)$	$= .17E-9$	
	μ	$= 1$	
	$I(x^*)$	$= (1)$	
	u^*_{max}/u^*_{min}	$= 3.0493/3.0493 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 2.86/2.86 = 1$	

PROBLEM:	12
CLASSIFICATION:	QQR-T1-3
SOURCE:	Mine e.al. [46]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = .5x_1^2 + x_2^2 - x_1x_2 - 7x_1 - 7x_2$$

CONSTRAINTS:

$$25 - 4x_1^2 - x_2^2 \geq 0$$

START:	x_o	= (0 , 0)	(feasible)
	$f(x_o)$	= 0	
SOLUTION:	x^*	= (2 , 3)	
	$f(x^*)$	= -30	
	$r(x^*)$	= 0	
	$e(x^*)$	= .81E-10	
	μ	= 1	
	$I(x^*)$	= (1)	
	u^*_{max}/u^*_{min}	= .5000/.5000 = 1	
	$\lambda^*_{max}/\lambda^*_{min}$	= 3.90/3.90 = 1	

PROBLEM:	13
CLASSIFICATION:	QPR-T1-1
SOURCE:	Betts [8], Kuhn, Tucker [38]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 2$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 2)^2 + x_2^2$$

CONSTRAINTS:

$$(1 - x_1)^3 - x_2 \geq 0$$

$$0 \leq x_1$$

$$0 \leq x_2$$

START:	x_o	$= (-2 , -2)$	(not feasible)
	$f(x_o)$	$= 20$	
SOLUTION:	x^*	$= (1 , 0)$	
	$f(x^*)$	$= 1$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 2$	(constraint qualification not satisfied)
	μ	$= 2$	
	$I(x^*)$	$= (1,3)$	
	u^*_{max}/u^*_{min}	$= 0/0$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= -$	

PROBLEM:	14
CLASSIFICATION:	QQR-T1-4
SOURCE:	Bracken, McCormick [13], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 1(1)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 2)^2 + (x_2 - 1)^2$$

CONSTRAINTS:

$$-.25x_1^2 - x_2^2 + 1 \geq 0$$

$$x_1 - 2x_2 + 1 = 0$$

START:	x_o	$= (2 , 2)$	(not feasible)
	$f(x_o)$	$= 1$	
SOLUTION:	x^*	$= (.5(\sqrt{7} - 1) , .25(\sqrt{7} + 1))$	
	$f(x^*)$	$= 9 - 2.875\sqrt{7}$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 1$	
	$I(x^*)$	$= (1)$	
	u^*_{max}/u^*_{min}	$= 1.8466/1.5945 = 1.15$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= -$	

PROBLEM:	15
CLASSIFICATION:	PQR-T1-1
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 1$

OBJECTIVE FUNCTION:

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

CONSTRAINTS:

$$x_1 x_2 - 1 \geq 0$$

$$x_1 + x_2^2 \geq 0$$

$$x_1 \leq .5$$

START:	x_o	= (-2 , 1)	(not feasible)
	$f(x_o)$	= 909	
SOLUTION:	x^*	= (.5 , 2)	
	$f(x^*)$	= 306.5	
	$r(x^*)$	= 0	
	$e(x^*)$	= 0	
	μ	= 2	
	$I(x^*)$	= (1 , 3)	
	u^*_{max}/u^*_{min}	= 1751/700 = 2.50	
	$\lambda^*_{max}/\lambda^*_{min}$	= -	

PROBLEM:	16
CLASSIFICATION:	PQR-T1-2
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 3$

OBJECTIVE FUNCTION:

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

CONSTRAINTS:

$$x_1 + x_2^2 \geq 0$$

$$x_1^2 + x_2 \geq 0$$

$$-.5 \leq x_1 \leq .5$$

$$x_2 \leq 1$$

START:	x_o	$= (-2 , 1)$	(not feasible)
	$f(x_o)$	$= 909$	
SOLUTION:	x^*	$= (.5 , .25)$	
	$f(x^*)$	$= .25$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 1$	
	$I(x^*)$	$= (4)$	
	u^*_{max}/u^*_{min}	$= 1.0000/1.0000 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 200/200 = 1$	

PROBLEM:	17
CLASSIFICATION:	PQR-T1-3
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 3$

OBJECTIVE FUNCTION:

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

CONSTRAINTS:

$$x_2^2 - x_1 \geq 0$$

$$x_1^2 - x_2 \geq 0$$

$$-.5 \leq x_1 \leq .5$$

$$x_2 \leq 1$$

START:	x_o	= (-2 , 1)	(not feasible)
	$f(x_o)$	= 909	

SOLUTION:	x^*	= (0 , 0)
	$f(x^*)$	= 1
	$r(x^*)$	= 0
	$e(x^*)$	= 0
	μ	= 2
	$I(x^*)$	= (1 , 2)
	u^*_{max}/u^*_{min}	= 2.0000/0
	$\lambda^*_{max}/\lambda^*_{min}$	= -

PROBLEM:	18
CLASSIFICATION:	QQR-T1-5
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = .01x_1^2 + x_2^2$$

CONSTRAINTS:

$$x_1x_2 - 25 \geq 0$$

$$x_1^2 + x_2^2 - 25 \geq 0$$

$$2 \leq x_1 \leq 50$$

$$0 \leq x_2 \leq 50$$

START:	x_o	$= (2 , 2)$	(not feasible)
	$f(x_o)$	$= 4.04$	
SOLUTION:	x^*	$= (\sqrt{250} , \sqrt{2.5})$	
	$f(x^*)$	$= 5$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= .24E-9$	
	μ	$= 1$	
	$I(x^*)$	$= (1)$	
	u^*_{max}/u^*_{min}	$= .2000/.2000 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= .079/.079 = 1$	

PROBLEM:	19
CLASSIFICATION:	PQR-T1-4
SOURCE:	Betts [8], Gould [27]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 10)^3 + (x_2 - 20)^3$$

CONSTRAINTS:

$$(x_1 - 5)^2 + (x_2 - 5)^2 - 100 \geq 0$$

$$-(x_2 - 5)^2 - (x_1 - 6)^2 + 82.81 \geq 0$$

$$13 \leq x_1 \leq 100$$

$$0 \leq x_2 \leq 100$$

START:	x_o	= (20.1 , 5.84)	(not feasible)
	$f(x_o)$	= -1808.858296	
SOLUTION:	x^*	= (14.095 , .84296079)	
	$f(x^*)$	= -6961.81381	
	$r(x^*)$	= 0	
	$e(x^*)$	= 0	
	μ	= 2	
	$I(x^*)$	= (1 , 2)	
	u^*_{max}/u^*_{min}	= 1229.5/1097.1 = 1.12	
	$\lambda^*_{max}/\lambda^*_{min}$	= -	

PROBLEM:	20
CLASSIFICATION:	PQR-T1-5
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 3$, $m-m_1 = 0$, $b = 2$

OBJECTIVE FUNCTION:

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

CONSTRAINTS:

$$x_1 + x_2^2 \geq 0$$

$$x_1^2 + x_2 \geq 0$$

$$x_1^2 + x_2^2 - 1 \geq 0$$

$$-.5 \leq x_1 \leq .5$$

START:	x_o	$= (-2 , 1)$	(not feasible)
	$f(x_o)$	$= 909$	
SOLUTION:	x^*	$= (.5 , .5\sqrt{3})$	
	$f(x^*)$	$= 81.5 - 25\sqrt{3}$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 2$	
	$I(x^*)$	$= (3 , 5)$	
	u^*_{max}/u^*_{min}	$= 195.34/71.132 = 2.75$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= -$	

PROBLEM:	21
CLASSIFICATION:	QLR-T1-1
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 1(1)$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = .01x_1^2 + x_2^2 - 100$$

CONSTRAINTS:

$$10x_1 - x_2 - 10 \geq 0$$

$$2 \leq x_1 \leq 50$$

$$-50 \leq x_2 \leq 50$$

START:	x_o	= $(-1 , -1)$	(not feasible)
	$f(x_o)$	= -98.99	
SOLUTION:	x^*	= $(2 , 0)$	
	$f(x^*)$	= -99.96	
	$r(x^*)$	= 0	
	$e(x^*)$	= 0	
	μ	= 1	
	$I(x^*)$	= (2)	
	u^*_{max}/u^*_{min}	= $.04/.04 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	= $-$	

PROBLEM:	22
CLASSIFICATION:	QQR-T1-6
SOURCE:	Bracken, McCormick [13], Himmelblau [29], Sheela [57]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 2(1)$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 2)^2 + (x_2 - 1)^2$$

CONSTRAINTS:

$$-x_1 - x_2 + 2 \geq 0$$

$$-x_1^2 + x_2 \geq 0$$

START:	x_o	=	(2 , 2) (not feasible)
	$f(x_o)$	=	1
SOLUTION:	x^*	=	(1 , 1)
	$f(x^*)$	=	1
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	2
	$I(x^*)$	=	(1 , 2)
	u^*_{max}/u^*_{min}	=	.6666/.6666 = 1
	$\lambda^*_{max}/\lambda^*_{min}$	=	-

PROBLEM:	23
CLASSIFICATION:	QQR-T1-7
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 5(1)$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = x_1^2 + x_2^2$$

CONSTRAINTS:

$$x_1 + x_2 - 1 \geq 0$$

$$x_1^2 + x_2^2 - 1 \geq 0$$

$$9x_1^2 + x_2^2 - 9 \geq 0$$

$$x_1^2 - x_2 \geq 0$$

$$x_2^2 - x_1 \geq 0$$

$$-50 \leq x_i \leq 50 \quad , \quad i=1,2$$

START:	x_o	$= (3 , 1)$	(not feasible)
	$f(x_o)$	$= 10$	

SOLUTION:	x^*	$= (1 , 1)$
	$f(x^*)$	$= 2$
	$r(x^*)$	$= 0$
	$e(x^*)$	$= 0$
	μ	$= 2$
	$I(x^*)$	$= (4 , 5)$
	u^*_{max}/u^*_{min}	$= 2/2 = 1$
	$\lambda^*_{max}/\lambda^*_{min}$	$= -$

PROBLEM:	24
CLASSIFICATION:	PLR-T1-1
SOURCE:	Betts [8], Box [12]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 3(3)$, $m-m_1 = 0$, $b = 2$

OBJECTIVE FUNCTION:

$$f(x) = \frac{1}{27\sqrt{3}} ((x_1 - 3)^2 - 9)\, x_2^{\,3}$$

CONSTRAINTS:

$$x_1/\sqrt{3} - x_2 \geq 0$$

$$x_1 + \sqrt{3}x_2 \geq 0$$

$$-x_1 - \sqrt{3}x_2 + 6 \geq 0$$

$$0 \leq x_1$$

$$0 \leq x_2$$

START:	x_o	$= (1 , .5)$	(feasible).
	$f(x_o)$	$= -.01336459$	

SOLUTION:	x^*	$= (3 , \sqrt{3})$
	$f(x^*)$	$= -1$
	$r(x^*)$	$= 0$
	$e(x^*)$	$= 0$
	μ	$= 2$
	$I(x^*)$	$= (1 , 3)$
	u^*_{max}/u^*_{min}	$= .86603/.5 = 1.73$
	$\lambda^*_{max}/\lambda^*_{min}$	$= -$

PROBLEM:	25
CLASSIFICATION:	SBR-T1-1
SOURCE:	Holzmann [32], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{i=1}^{99} (f_i(x))^2$$

$$f_i(x) = -.01i + \exp(-\frac{1}{x_1}(u_i - x_2)^{x_3})$$

$$u_i = 25 + (-50 \ln(.01i))^{2/3}$$

$$i = 1,\ldots,99$$

CONSTRAINTS:

$$.1 \le x_1 \le 100$$

$$0 \le x_2 \le 25.6$$

$$0 \le x_3 \le 5$$

START: $x_o = (100 , 12.5 , 3)$ $f(x_o) = 32.835$ (feasible)

SOLUTION: $f(x^*) = 0$

$x^* = (50 , 25 , 1.5)$

$r(x^*) = 0$ $e(x^*) = -$

$\mu = 0$ $I(x^*) = -$

$u^*_{max}/u^*_{min} = -$

$\lambda^*_{max}/\lambda^*_{min} = 94.7/.14E-4 = .70E7$

PROBLEM:	26
CLASSIFICATION:	PPR-T1-1
SOURCE:	Huang, Aggerwal [34], Miele e.al. [43]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - x_2)^2 + (x_2 - x_3)^4$$

CONSTRAINTS:

$$(1 + x_2^2)x_1 + x_3^4 - 3 = 0$$

START:	x_o	=	(-2.6 , 2 , 2) (feasible)
	$f(x_o)$	=	21.16
SOLUTION:	x^*	=	(1 , 1 , 1) , (a , a , a)
	$f(x^*)$	=	0
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	0
	$\lambda^*_{max}/\lambda^*_{min}$	=	4/0

$a = \sqrt[3]{\alpha-\beta} - \sqrt[3]{\alpha+\beta} - 2/3$

$\alpha = \sqrt{139/108}$

$\beta = 61/54$

PROBLEM:	27
CLASSIFICATION:	PQR-T1-6
SOURCE:	Miele e.al. [44,45]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = .01(x_1 - 1)^2 + (x_2 - x_1^2)^2$$

CONSTRAINTS:

$$x_1 + x_3^2 + 1 = 0$$

START:	x_o	= (2 , 2 , 2)	(not feasible)
	$f(x_o)$	= 4.01	
SOLUTION:	x^*	= (-1 , 1 , 0)	
	$f(x^*)$	= .04	
	$r(x^*)$	= 0	
	$e(x^*)$	= 0	
	μ	= 0	
	$I(x^*)$	= -	
	u^*_{max}/u^*_{min}	= .04/.04 = 1	
	$\lambda^*_{max}/\lambda^*_{min}$	= 2/.08 = 25	

PROBLEM:	28
CLASSIFICATION:	QLR-T1-2
SOURCE:	Huang, Aggerwal [34]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1(1)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 + x_2)^2 + (x_2 + x_3)^2$$

CONSTRAINTS:

$$x_1 + 2x_2 + 3x_3 - 1 = 0$$

START:	x_o	$= (-4 , 1 , 1)$	(feasible)
	$f(x_o)$	$= 13$	
SOLUTION:	x^*	$= (.5 , -.5 , .5)$	
	$f(x^*)$	$= 0$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 0$	
	$I(x^*)$	$= -$	
	u^*_{max}/u^*_{min}	$= 0$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 2.72/.42 = 6.45$	

PROBLEM:	29
CLASSIFICATION:	PQR-T1-7
SOURCE:	Biggs [10]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = -x_1x_2x_3$$

CONSTRAINTS:

$$-x_1^2 - 2x_2^2 - 4x_3^2 + 48 \geq 0$$

START:	x_o	= (1 , 1 , 1)	(feasible)
	$f(x_o)$	= -1	
SOLUTION:	x^*	= (a,b,c) , (a,-b,-c) ,	(-a,b,-c) ,
	$f(x^*)$	= $-16\sqrt{2}$	(-a,-b,c)
	$r(x^*)$	= 0	a = 4
	$e(x^*)$	= .19E-9	b = $2\sqrt{2}$
	μ	= 1	c = 2
	$I(x^*)$	= (1)	
	u^*_{max}/u^*_{min}	= .7071/.7071 = 1	
	$\lambda^*_{max}/\lambda^*_{min}$	= 7.79/3.52 = 2.22	

PROBLEM:	30
CLASSIFICATION:	QQR-T1-8
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = x_1^2 + x_2^2 + x_3^2$$

CONSTRAINTS:

$$x_1^2 + x_2^2 - 1 \geq 0$$

$$1 \leq x_1 \leq 10$$

$$-10 \leq x_2 \leq 10$$

$$-10 \leq x_3 \leq 10$$

START:	x_o	$= (1 , 1 , 1)$	(feasible)
	$f(x_o)$	$= 3$	
SOLUTION:	x^*	$= (1 , 0 , 0)$	
	$f(x^*)$	$= 1$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 2$	
	$I(x^*)$	$= (1 , 2)$	
	u^*_{max}/u^*_{min}	$= 1/0$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 2/2 = 1$	

PROBLEM:	31
CLASSIFICATION:	QQR-T1-9
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = 9x_1^2 + x_2^2 + 9x_3^2$$

CONSTRAINTS:

$$x_1x_2 - 1 \geq 0$$

$$-10 \leq x_1 \leq 10$$

$$1 \leq x_2 \leq 10$$

$$-10 \leq x_3 \leq 1$$

START:	x_o	$= (1 , 1 , 1)$	(feasible)
	$f(x_o)$	$= 19$	
SOLUTION:	x^*	$= (1/\sqrt{3} , \sqrt{3} , 0)$	
	$f(x^*)$	$= 6$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= .57E-10$	
	μ	$= 1$	
	$I(x^*)$	$= (1)$	
	u^*_{max}/u^*_{min}	$= 6/6 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 18/7.2 = 2.5$	

PROBLEM:	32
CLASSIFICATION:	QPR-T1-2
SOURCE:	Evtushenko [25]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 1(1)$, $b = 3$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 + 3x_2 + x_3)^2 + 4(x_1 - x_2)^2$$

CONSTRAINTS:

$$6x_2 + 4x_3 - x_1^3 - 3 \geq 0$$

$$1 - x_1 - x_2 - x_3 = 0$$

$$0 \leq x_i \quad , \quad i = 1,2,3$$

START:	x_o	$= (.1 , .7 , .2)$	(feasible)
	$f(x_o)$	$= 7.2$	
SOLUTION:	x^*	$= (0 , 0 , 1)$	
	$f(x^*)$	$= 1$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 2$	
	$I(x^*)$	$= (2 , 3)$	
	u^*_{max}/u^*_{min}	$= 4/0$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= -$	

PROBLEM:	33
CLASSIFICATION:	PQR-T1-8
SOURCE:	Beltrami [6], Hartmann [28]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 1)(x_1 - 2)(x_1 - 3) + x_3$$

CONSTRAINTS:

$$x_3^2 - x_2^2 - x_1^2 \geq 0$$

$$x_1^2 + x_2^2 + x_3^2 - 4 \geq 0$$

$$0 \leq x_1$$

$$0 \leq x_2$$

$$0 \leq x_3 \leq 5$$

START: $x_o = (0 , 0 , 3)$ (feasible)

$f(x_o) = -3$

SOLUTION:

$x^* = (0 , \sqrt{2} , \sqrt{2})$

$f(x^*) = \sqrt{2} - 6$

$r(x^*) = 0$

$e(x^*) = 0$

$\mu = 3$

$I(x^*) = (1 , 2 , 3)$

$u^*_{max}/u^*_{min} = 11/.17678 = 62.23$

$\lambda^*_{max}/\lambda^*_{min} = -$

PROBLEM:	34
CLASSIFICATION:	LGR-T1-1
SOURCE:	Eckhardt [24]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = -x_1$$

CONSTRAINTS:

$$x_2 - \exp(x_1) \geq 0$$

$$x_3 - \exp(x_2) \geq 0$$

$$0 \leq x_1 \leq 100$$

$$0 \leq x_2 \leq 100$$

$$0 \leq x_3 \leq 10$$

START:	x_o	$= (0 , 1.05 , 2.9)$	(feasible)
	$f(x_o)$	$= 0$	
SOLUTION:	x^*	$= (\ln(\ln 10) , \ln 10 , 10)$	
	$f(x^*)$	$= -\ln(\ln 10)$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 3$	
	$I(x^*)$	$= (1 , 2 , 8)$	
	u^*_{max}/u^*_{min}	$= .4343/.04343 = 10$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= -$	

PROBLEM:	35 (Beale's problem)
CLASSIFICATION:	QLR-T1-3
SOURCE:	Asaadi [1], Charalambous [18], Dimitru [23], Sheela [57]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 1(1)$, $m-m_1 = 0$, $b = 3$

OBJECTIVE FUNCTION:

$$f(x) = 9 - 8x_1 - 6x_2 - 4x_3 + 2x_1^2 + 2x_2^2 + x_3^2 + 2x_1x_2 + 2x_1x_3$$

CONSTRAINTS:

$$3 - x_1 - x_2 - 2x_3 \geq 0$$

$$0 \leq x_i \quad , \quad i=1,2,3$$

START:	x_o	= (.5 , .5 , .5)	(feasible)
	$f(x_o)$	= 2.25	
SOLUTION:	x^*	= (4/3 , 7/9 , 4/9)	
	$f(x^*)$	= 1/9	
	$r(x^*)$	= 0	
	$e(x^*)$	= .49E-10	
	μ	= 1	
	$I(x^*)$	= (1)	
	u^*_{max}/u^*_{min}	= .2222/.2222 = 1	
	$\lambda^*_{max}/\lambda^*_{min}$	= 3.72/1.61 = 2.31	

PROBLEM:	36
CLASSIFICATION:	PLR-T1-2
SOURCE:	Biggs [10]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 1(1)$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = -x_1x_2x_3$$

CONSTRAINTS:

$$72 - x_1 - 2x_2 - 2x_3 \geq 0$$

$$0 \leq x_1 \leq 20$$

$$0 \leq x_2 \leq 11$$

$$0 \leq x_3 \leq 42$$

START:	x_o	= (10 , 10 , 10)	(feasible)
	$f(x_o)$	= -1000	
SOLUTION:	x^*	= (20 , 11 , 15)	
	$f(x^*)$	= -3300	
	$r(x^*)$	= 0	
	$e(x^*)$	= 0	
	μ	= 3	
	$I(x^*)$	= (1 , 5 , 6)	
	u^*_{max}/u^*_{min}	= 110/55 = 2	
	$\lambda^*_{max}/\lambda^*_{min}$	= -	

PROBLEM:	37
CLASSIFICATION:	PLR-T1-3
SOURCE:	Betts [8], Box [12]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 2(2)$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = -x_1x_2x_3$$

CONSTRAINTS:

$$72 - x_1 - 2x_2 - 2x_3 \geq 0$$

$$x_1 + 2x_2 + 2x_3 \geq 0$$

$$0 \leq x_i \leq 42 \quad , \quad i=1,2,3$$

START:	x_o	$= (10 , 10 , 10)$	(feasible)
	$f(x_o)$	$= -1000$	
SOLUTION:	x^*	$= (24 , 12 , 12)$	
	$f(x^*)$	$= -3456$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 1$	
	$I(x^*)$	$= (1)$	
	u^*_{max}/u^*_{min}	$= 144/144 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 24/8 = 3$	

PROBLEM:	38 (Colville No.4)
CLASSIFICATION:	PBR-T1-4
SOURCE:	Colville [20], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2 + 90(x_4 - x_3^2)^2 + (1-x_3)^2$$
$$+ 10.1((x_2 - 1)^2 + (x_4 - 1)^2) + 19.8(x_2 - 1)(x_4 - 1)$$

CONSTRAINTS:

$$-10 \le x_i \le 10 \quad , \quad i=1,\ldots,4$$

START:	x_o	=	(-3 , -1 , -3 , -1) (feasible)
	$f(x_o)$	=	19192
SOLUTION:	x^*	=	(1 , 1 , 1 , 1)
	$f(x^*)$	=	0
	$r(x^*)$	=	0
	$e(x^*)$	=	-
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	-
	$\lambda^*_{max}/\lambda^*_{min}$	=	.10E4/.72 = .14E4

PROBLEM:	39
CLASSIFICATION:	LPR-T1-1
SOURCE:	Miele e.al. [44,45]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = -x_1$$

CONSTRAINTS:

$$x_2 - x_1^3 - x_3^2 = 0$$

$$x_1^2 - x_2 - x_4^2 = 0$$

START:	x_o	$= (2, 2, 2, 2)$	(not feasible)
	$f(x_o)$	$= -2$	
SOLUTION:	x^*	$= (1, 1, 0, 0)$	
	$f(x^*)$	$= -1$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 0$	
	$I(x^*)$	$= -$	
	u^*_{max}/u^*_{min}	$= 1/1 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 2/2 = 1$	

PROBLEM:	40
CLASSIFICATION:	PPR-T1-2
SOURCE:	Beltrami [6], Indusi [35]
NUMBER OF VARIABLES:	n =4
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = -x_1x_2x_3x_4$$

CONSTRAINTS:

$$x_1^3 + x_2^2 - 1 = 0$$

$$x_1^2x_4 - x_3 = 0$$

$$x_4^2 - x_2 = 0$$

START:	x_o	=	(.8 , .8 , .8 , .8) (not feasible)
	$f(x_o)$	=	-.4096
SOLUTION:	x^*	=	$(2^a , 2^{2b} , (-1)^i 2^c , (-1)^i 2^b)$
	$f(x^*)$	=	-.25 i=1,2
	$r(x^*)$	=	0 a = -1/3
	$e(x^*)$	=	.80E-11 b = -1/4
	μ	=	0 c = -11/12
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	.5/.3536 = 1.41
	$\lambda^*_{max}/\lambda^*_{min}$	=	1.74/1.74 = 1

PROBLEM:	41
CLASSIFICATION:	PLR-T1-4
SOURCE:	Betts [8], Miele e.al. [42]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1(1)$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = 2 - x_1 x_2 x_3$$

CONSTRAINTS:

$$x_1 + 2x_2 + 2x_3 - x_4 = 0$$

$$0 \le x_i \le 1 \quad , \quad i=1,2,3$$

$$0 \le x_4 \le 2$$

START:	x_o	$= (2 , 2 , 2 , 2)$	(not feasible)
	$f(x_o)$	$= -6$	
SOLUTION:	x^*	$= (2/3 , 1/3 , 1/3 , 2)$	
	$f(x^*)$	$= 52/27$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= .13E-10$	
	μ	$= 1$	
	$I(x^*)$	$= (8)$	
	u^*_{max}/u^*_{min}	$= .1111/.1111 = 1$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= .67/.22 = 3$	

PROBLEM:	42
CLASSIFICATION:	QQR-T1-10
SOURCE:	Brusch [14]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2(1)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 1)^2 + (x_2 - 2)^2 + (x_3 - 3)^2 + (x_4 - 4)^2$$

CONSTRAINTS:

$$x_1 - 2 = 0$$

$$x_3^2 + x_4^2 - 2 = 0$$

START:	x_o	$= (1 , 1 , 1 , 1)$	(not feasible)
	$f(x_o)$	$= 14$	
SOLUTION:	x^*	$= (2 , 2 , .6\sqrt{2} , .8\sqrt{2})$	
	$f(x^*)$	$= 28 - 10\sqrt{2}$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= .2E-23$	
	μ	$= 0$	
	$I(x^*)$	$= -$	
	u^*_{max}/u^*_{min}	$= 2.5355/2.0000 = 1.26$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 7.07/2.00 = 3.54$	

PROBLEM:	43 (Rosen-Suzuki)
CLASSIFICATION:	QQR-T1-11
SOURCE: Betts [8], Charalambous [18], Gould [27], Sheela [57]	
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 3$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = x_1^2 + x_2^2 + 2x_3^2 + x_4^2 - 5x_1 - 5x_2 - 21x_3 + 7x_4$$

CONSTRAINTS:

$$8 - x_1^2 - x_2^2 - x_3^2 - x_4^2 - x_1 + x_2 - x_3 + x_4 \geq 0$$

$$10 - x_1^2 - 2x_2^2 - x_3^2 - 2x_4^2 + x_1 + x_4 \geq 0$$

$$5 - 2x_1^2 - x_2^2 - x_3^2 - 2x_1 + x_2 + x_4 \geq 0$$

START:	x_o	$= (0 , 0 , 0 , 0)$	(feasible)
	$f(x_o)$	$= 0$	
SOLUTION:	x^*	$= (0 , 1 , 2 , -1)$	
	$f(x^*)$	$= -44$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= .21E-9$	
	μ	$= 2$	
	$I(x^*)$	$= (1 , 3)$	
	u^*_{max}/u^*_{min}	$= 2/1 = 2$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 9/8.07 = 1.12$	

PROBLEM:	44
CLASSIFICATION:	QLR-T1-4
SOURCE:	Konno [37]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 6(6)$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = x_1 - x_2 - x_3 - x_1x_3 + x_1x_4 + x_2x_3 - x_2x_4$$

CONSTRAINTS:

$$8 - x_1 - 2x_2 \geq 0$$

$$12 - 4x_1 - x_2 \geq 0$$

$$12 - 3x_1 - 4x_2 \geq 0$$

$$8 - 2x_3 - x_4 \geq 0$$

$$8 - x_3 - 2x_4 \geq 0$$

$$5 - x_3 - x_4 \geq 0 \quad , \quad 0 \leq x_i \quad , \quad i=1,\ldots,4$$

START:	x_o	=	$(0 , 0 , 0 , 0)$ (feasible)
	$f(x_o)$	=	0
SOLUTION:	x^*	=	$(0 , 3 , 0 , 4)$
	$f(x^*)$	=	-15
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	4
	$I(x^*)$	=	$(3 , 5 , 7 , 9)$
	u^*_{max}/u^*_{min}	=	$8.75/1.25 = 7$
	$\lambda^*_{max}/\lambda^*_{min}$	=	-

PROBLEM:	45
CLASSIFICATION:	PBR-T1-5
SOURCE:	Betts [8], Miele e.al. [42]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 10$

OBJECTIVE FUNCTION:

$$f(x) = 2 - \frac{1}{120} x_1 x_2 x_3 x_4 x_5$$

CONSTRAINTS:

$$0 \leq x_i \leq i \quad , \quad i=1,\ldots,5$$

START:	x_o	= (2 , 2 , 2 , 2)	(not feasible)
	$f(x_o)$	= 26/15	
SOLUTION:	x^*	= (1 , 2 , 3 , 4 , 5)	
	$f(x^*)$	= 1	
	$r(x^*)$	= 0	
	$e(x^*)$	= 0	
	μ	= 5	
	$I(x^*)$	= (6 , 7 , 8 , 9 , 10)	
	u^*_{max}/u^*_{min}	= 1/.2 = 5	
	$\lambda^*_{max}/\lambda^*_{min}$	= -	

PROBLEM:	46
CLASSIFICATION:	PGR-T1-1
SOURCE:	Huang, Aggerwal [34], Miele e.al. [43]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - x_2)^2 + (x_3 - 1)^2 + (x_4 - 1)^4 + (x_5 - 1)^6$$

CONSTRAINTS:

$$x_1^2 x_4 + \sin(x_4 - x_5) - 1 = 0$$

$$x_2 + x_3^4 x_4^2 - 2 = 0$$

START:	x_o	$= (.5\sqrt{2}\ ,\ 1.75\ ,\ .5\ ,\ 2\ ,\ 2)$	(feasible)
	$f(x_o)$	$= 5.8125 - 1.75\sqrt{2}$	
SOLUTION:	x^*	$= (1\ ,\ 1\ ,\ 1\ ,\ 1\ ,\ 1)$	
	$f(x^*)$	$= 0$	
	$r(x^*)$	$= 0$	
	$e(x^*)$	$= 0$	
	μ	$= 0$	
	$I(x^*)$	$= -$	
	u^*_{max}/u^*_{min}	$= 0/0$	
	$\lambda^*_{max}/\lambda^*_{min}$	$= 2.97/.74E-7 = .40E8$	

PROBLEM:	47
CLASSIFICATION:	PPR-T1-3
SOURCE:	Huang, Aggerwal [34], Miele e.al. [43]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - x_2)^2 + (x_2 - x_3)^3 + (x_3 - x_4)^4 + (x_4 - x_5)^4$$

CONSTRAINTS:

$$x_1 + x_2^2 + x_3^3 - 3 = 0$$

$$x_2 - x_3^2 + x_4 - 1 = 0$$

$$x_1 x_5 - 1 = 0$$

START:	x_o	=	$(2 , \sqrt{2} , -1 , 2-\sqrt{2} , .5)$ (feasible)
	$f(x_o)$	=	12.4954368
SOLUTION:	x^*	=	$(1 , 1 , 1 , 1 , 1)$
	$f(x^*)$	=	0
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	0/0
	$\lambda^*_{max}/\lambda^*_{min}$	=	2.08/.53 = 3.92

PROBLEM:	48
CLASSIFICATION:	QLR-T1-5
SOURCE:	Huang, Aggerwal [34], Miele e.al. [43]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2(2)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 1)^2 + (x_2 - x_3)^2 + (x_4 - x_5)^2$$

CONSTRAINTS:

$$x_1 + x_2 + x_3 + x_4 + x_5 - 5 = 0$$

$$x_3 - 2(x_4 + x_5) + 3 = 0$$

START:	x_o	=	(3 , 5 , -3 , 2 , -2) (feasible)
	$f(x_o)$	=	84
SOLUTION:	x^*	=	(1 , 1 , 1 , 1 , 1)
	$f(x^*)$	=	0
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	0/0
	$\lambda^*_{max}/\lambda^*_{min}$	=	4/1.49 = 2.69

PROBLEM:	49
CLASSIFICATION:	PLR-T1-5
SOURCE:	Huang, Aggerwal [34]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2(2)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - x_2)^2 + (x_3 - 1)^2 + (x_4 - 1)^4 + (x_5 - 1)^6$$

CONSTRAINTS:

$$x_1 + x_2 + x_3 + 4x_4 - 7 = 0$$

$$x_3 + 5x_5 - 6 = 0$$

START:	x_o	=	(10 , 7 , 2 , -3 , .8) (feasible)
	$f(x_o)$	=	266
SOLUTION:	x^*	=	(1 , 1 , 1 , 1 , 1)
	$f(x^*)$	=	0
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	0/0
	$\lambda^*_{max}/\lambda^*_{min}$	=	4/.70E-10 = .57E11

PROBLEM:	50
CLASSIFICATION:	PLR-T1-6
SOURCE:	Huang, Aggerwal [34]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3(3)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_3 - x_4)^4 + (x_4 - x_5)^2$$

CONSTRAINTS:

$$x_1 + 2x_2 + 3x_3 - 6 = 0$$

$$x_2 + 2x_3 + 3x_4 - 6 = 0$$

$$x_3 + 2x_4 + 3x_5 - 6 = 0$$

START:	x_o	=	(35 , -31 , 11 , 5 , -5) (feasible)
	$f(x_o)$	=	17416
SOLUTION:	x^*	=	(1 , 1 , 1 , 1 , 1)
	$f(x^*)$	=	0
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	0/0
	$\lambda^*_{max}/\lambda^*_{min}$	=	5.89/1.64 = 3.6

PROBLEM:	51
CLASSIFICATION:	QLR-T1-6
SOURCE:	Huang, Aggerwal [34]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3(3)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - x_2)^2 + (x_2 + x_3 - 2)^2 + (x_4 - 1)^2 + (x_5 - 1)^2$$

CONSTRAINTS:

$$x_1 + 3x_2 - 4 = 0$$

$$x_3 + x_4 - 2x_5 = 0$$

$$x_2 - x_5 = 0$$

START:	x_o	=	(2.5 , .5 , 2 , -1 , .5) (feasible)
	$f(x_o)$	=	8.5
SOLUTION:	x^*	=	(1 , 1 , 1 , 1 , 1)
	$f(x^*)$	=	0
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	0/0
	$\lambda^*_{max}/\lambda^*_{min}$	=	3.49/1.90 = 1.84

PROBLEM:	52
CLASSIFICATION:	QLR-T1-7
SOURCE:	Miele e.al. [44,45]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3(3)$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (4x_1 - x_2)^2 + (x_2 + x_3 - 2)^2 + (x_4 - 1)^2 + (x_5 - 1)^2$$

CONSTRAINTS:

$$x_1 + 3x_2 = 0$$

$$x_3 + x_4 - 2x_5 = 0$$

$$x_2 - x_5 = 0$$

START:	x_o	=	(2 , 2 , 2 , 2 , 2) (not feasible)
	$f(x_o)$	=	42
SOLUTION:	x^*	=	(-33 , 11 , 180 , -158 , 11)/349
	$f(x^*)$	=	1859/349
	$r(x^*)$	=	0
	$e(x^*)$	=	.14E-9
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	7.7479/2.9054 = 2.6667
	$\lambda^*_{max}/\lambda^*_{min}$	=	26.93/1.99 = 13.51

PROBLEM:	53
CLASSIFICATION:	QLR-T1-8
SOURCE:	Betts [8], Miele e.al. [42,43]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3(3)$, $b = 10$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - x_2)^2 + (x_2 + x_3 - 2)^2 + (x_4 - 1)^2 + (x_5 - 1)^2$$

CONSTRAINTS:

$$x_1 + 3x_2 = 0$$

$$x_3 + x_4 - 2x_5 = 0$$

$$x_2 - x_5 = 0$$

$$-10 \le x_i \le 10 \quad , \quad i=1,\ldots,5$$

START:	x_o	=	(2 , 2 , 2 , 2 , 2) (not feasible)
	$f(x_o)$	=	6
SOLUTION:	x^*	=	(-33 , 11 , 27 , -5 , 11)/43
	$f(x^*)$	=	176/43
	$r(x^*)$	=	0
	$e(x^*)$	=	.28E-9
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	5.9535/2.0465 = 1.84
	$\lambda^*_{max}/\lambda^*_{min}$	=	3.49/1.90 = 1.84

PROBLEM:	54
CLASSIFICATION:	GLR-T1-2
SOURCE:	Betts [8], Picket [50]
NUMBER OF VARIABLES:	$n = 6$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1(1)$, $b = 12$

OBJECTIVE FUNCTION:

$$f(x) = -\exp(-h(x)/2)$$

$$\begin{aligned} h(x) = {} & ((x_1 - 1.E6)^2/6.4E7 + (x_1 - 1.E4)(x_2 - 1)/2.E4 \\ & + (x_2 - 1)^2)(x_3 - 2.E6)^2/(.96 \cdot 4.9E13) \\ & + (x_4 - 10)^2/2.5E3 + (x_5 - 1.E-3)^2/2.5E-3 \\ & + (x_6 - 1.E8)^2/2.5E17 \end{aligned}$$

CONSTRAINTS:

$$x_1 + 4.E3x_2 - 1.76E4 = 0$$

$$0 \le x_1 \le 2.E4 \qquad -10 \le x_2 \le 10 \qquad 0 \le x_3 \le 1.E7$$

$$0 \le x_4 \le 20 \qquad -1 \le x_5 \le 1 \qquad 0 \le x_6 \le 2.E8$$

START:

x_o = (6E3 , 1.5 , 4E6 , 2 , 3E-3 , 5E7)

$f(x_o)$ = -.7651 (not feasible)

SOLUTION:

x^* = (91600/7,79/70,2E6,10,1E-3,1E8)

$f(x^*)$ = $-\exp(-27/280)$

$r(x^*)$ = 0

$e(x^*)$ = .20E-10

μ = 0

$I(x^*)$ = -

u^*_{max}/u^*_{min} = .4865E-4/.4865E-4 = 1

$\lambda^*_{max}/\lambda^*_{min}$ = 362.9/.36E-17 = .10E21

PROBLEM:	55
CLASSIFICATION:	GLR-T1-3
SOURCE:	Hsia [33]
NUMBER OF VARIABLES:	$n = 6$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 6(6)$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = x_1 + 2x_2 + 4x_5 + \exp(x_1 x_4)$$

CONSTRAINTS:

$$x_1 + 2x_2 + 5x_5 - 6 = 0$$

$$x_1 + x_2 + x_3 - 3 = 0$$

$$x_4 + x_5 + x_6 - 2 = 0$$

$$x_1 + x_4 - 1 = 0$$

$$x_2 + x_5 - 2 = 0$$

$$x_3 + x_6 - 2 = 0$$

$$0 \le x_1 \;,\; i=1,\ldots,6 \;,\; x_1 \le 1 \;,\; x_4 \le 1$$

START: $x_o = (1 , 2 , 0 , 0 , 0 , 2)$ (not feasible)

$f(x_o) = 6$

SOLUTION:

$x^* = (0 , 4/3 , 5/3 , 1 , 2/3 , 1/3)$

$f(x^*) = 19/3$

$r(x^*) = 0$

$e(x^*) = 0$

$\mu = 8$

$I(x^*) = (1 , 8)$

$u^*_{max}/u^*_{min} = -$

$\lambda^*_{max}/\lambda^*_{min} = -$

PROBLEM:	56
CLASSIFICATION:	PGR-T1-2
SOURCE:	Brusch [15]
NUMBER OF VARIABLES:	$n = 7$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 4$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = -x_1x_2x_3$$

CONSTRAINTS:

$$x_1 - 4.2\sin^2 x_4 = 0$$

$$x_2 - 4.2\sin^2 x_5 = 0$$

$$x_3 - 4.2\sin^2 x_6 = 0$$

$$x_1 + 2x_2 + 2x_3 - 7.2\sin^2 x_7 = 0$$

START:	x_o	=	.(1 , 1 , 1 , a , a , a , b)
	$f(x_o)$	=	-1 (feasible)
SOLUTION:	x^*	=	$(2.4, 1.2, 1.2, \pm c+j\pi, \pm d+k\pi, \pm d+l\pi, (r+.5)\pi)$
	$f(x^*)$	=	-3.456 $\quad a = \arcsin\sqrt{1/4.2}$
	$r(x^*)$	=	0 $\quad b = \arcsin\sqrt{5/7.2}$
	$e(x^*)$	=	.67E-10 $\quad c = \arcsin\sqrt{4/7}$
	μ	=	0 $\quad d = \arcsin\sqrt{2/7}$
	$I(x^*)$	=	- $\quad j,k,l,r = 0, \pm 1, \pm 2, \ldots$
	u^*_{max}/u^*_{min}	=	1.44/.68E-11 = .21E12
	$\lambda^*_{max}/\lambda^*_{min}$	=	20.74/.76 = 27.45

PROBLEM:	57
CLASSIFICATION:	SQR-P1-1
SOURCE:	Betts [8], Gould [27]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 2$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{i=1}^{44} f_i(x)^2$$

$$f_i(x) = b_i - x_1 - (.49 - x_1)\exp(-x_2(a_i - 8))$$

$$i=1,\ldots,44$$

a_i, b_i: cf. Appendix A

CONSTRAINTS:

$$.49x_2 - x_1x_2 - .09 \geq 0$$

$$.4 \leq x_1 \quad , \quad -4 \leq x_2$$

START:	x_o	= (.42 , 5)	(feasible)
	$f(x_o)$	= .030798602	
SOLUTION:	x^*	= (.419952675 , 1.284845629)	
	$f(x^*)$	= .02845966972	
	$r(x^*)$	= 0	
	$e(x^*)$	= .98E-7	
	μ	= 1	
	$I(x^*)$	= (1)	
	u^*_{max}/u^*_{min}	= .06671/.06671 = 1	
	$\lambda^*_{max}/\lambda^*_{min}$	= .23/.23 = 1	

PROBLEM:	59
CLASSIFICATION:	GQR-P1-1
SOURCE:	Barnes [3], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 2$
NUMBER OF CONSTRAINTS:	$m_1 = 3$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$
\begin{aligned}
f(x) = & -75.196 + 3.8112x_1 + .0020567x_1^3 - 1.0345E\text{-}5x_1^4 \\
& + 6.8306x_2 - .030234x_1x_2 + 1.28134E\text{-}3x_2x_1^2 \\
& + 2.266E\text{-}7x_1^4x_2 - .25645x_2^2 + .0034604x_2^3 - 1.3514E\text{-}5x_2^4 \\
& + 28.106/(x_2 + 1) + 5.2375E\text{-}6x_1^2x_2^2 + 6.3E\text{-}8x_1^3x_2^2 \\
& - 7E\text{-}10x_1^3x_2^3 - 3.405E\text{-}4x_1x_2^2 + 1.6638E\text{-}6x_1x_2^3 \\
& + 2.8673\exp(.0005x_1x_2) - 3.5256E\text{-}5x_1^3x_2
\end{aligned}
$$

CONSTRAINTS:

$$x_1x_2 - 700 \geq 0$$

$$x_2 - x_1^2/125 \geq 0 \qquad 0 \leq x_1 \leq 75$$

$$(x_2 - 50)^2 - 5(x_1 - 55) \geq 0 \qquad 0 \leq x_2 \leq 65$$

START: $x_o = (90 , 10)$ (not feasible)

$f(x_o) = 86.878639$

SOLUTION:

$x^* = (13.55010424 , 51.66018129)$

$f(x^*) = -7.804226324$

$r(x^*) = 0$

$e(x^*) = .27E\text{-}6$

$\mu = 1$

$I(x^*) = (1)$

$u^*_{max}/u^*_{min} = .01142/.01142 = 1$

$\lambda^*_{max}/\lambda^*_{min} = .13/.13 = 1$

PROBLEM:	60
CLASSIFICATION:	PPR-P1-1
SOURCE:	Betts [8], Miele e.al. [42,44]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 1)^2 + (x_1 - x_2)^2 + (x_2 - x_3)^4$$

CONSTRAINTS:

$$x_1(1 + x_2^2) + x_3^4 - 4 - 3\sqrt{2} = 0$$

$$-10 \le x_i \le 10 \quad , \quad i=1,2,3$$

START:	x_o	=	(2 , 2 , 2) (not feasible)
	$f(x_o)$	=	1
SOLUTION:	x^*	=	(1.104859024,1.196674194,1,535262257)
	$f(x^*)$	=	.03256820025
	$r(x^*)$	=	.23E-9
	$e(x^*)$	=	.38E-7
	μ	=	1
	$I(x^*)$	=	(1)
	u^*_{max}/u^*_{min}	=	.01073/.01073 = 1
	$\lambda^*_{max}/\lambda^*_{min}$	=	5.72/2.07 = 2.76

PROBLEM:	61
CLASSIFICATION:	QQR-P1-1
SOURCE:	Fletcher, Lill [26]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = 4x_1^2 + 2x_2^2 + 2x_3^2 - 33x_1 + 16x_2 - 24x_3$$

CONSTRAINTS:

$$3x_1 - 2x_2^2 - 7 = 0$$

$$4x_1 - x_3^2 - 11 = 0$$

START:	x_o	=	(0 , 0 , 0) (not feasible)
	$f(x_o)$	=	0
SOLUTION:	x^*	=	(5.326770157,-2.118998639,3.210464239)
	$f(x^*)$	=	-143.6461422
	$r(x^*)$	=	.29E-9
	$e(x^*)$	=	.21E-6
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	1.7378/.8877 = 1.96
	$\lambda^*_{max}/\lambda^*_{min}$	=	7.83/7.83 = 1

PROBLEM:	62
CLASSIFICATION:	GLR-P1-1
SOURCE:	Betts [8], Picket [50]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 1(1)$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = -32.174(255 \ln((x_1+x_2+x_3+.03)/(.09x_1+x_2+x_3+.03)) + 280 \ln((x_2+x_3+.03)/(.07x_2+x_3+.03)) + 290 \ln((x_3+.03)/(.13x_3+.03)))$$

CONSTRAINTS:

$$x_1 + x_2 + x_3 - 1 = 0$$

$$0 \le x_i \le 1 \quad , \quad i=1,2,3$$

START:	x_o	= (.7 , .2 , .1)	(feasible)
	$f(x_o)$	= -25698.3	

SOLUTION:	x^*	= (.6178126908,.328202223,.5398508606E-1)
	$f(x^*)$	= -26272.51448
	$r(x^*)$	= 0
	$e(x^*)$	= .20E-5
	μ	= 0
	$I(x^*)$	= -
	u^*_{max}/u^*_{min}	= 6387/6387 = 1
	$\lambda^*_{max}/\lambda^*_{min}$	= .32E6/.72E4 = 44.9

PROBLEM:	63
CLASSIFICATION:	QQR-P1-2
SOURCE:	Himmelblau [29], Paviani [48], Sheela [57]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2(1)$, $b = 3$

OBJECTIVE FUNCTION:

$$f(x) = 1000 - x_1^2 - 2x_2^2 - x_3^2 - x_1x_2 - x_1x_3$$

CONSTRAINTS:

$$8x_1 + 14x_2 + 7x_3 - 56 = 0$$

$$x_1^2 + x_2^2 + x_3^2 - 25 = 0$$

$$0 \leq x_i \quad , \quad i=1,2,3$$

START:	x_o	= (2 , 2 , 2)	(not feasible)
	$f(x_o)$	= 976	
SOLUTION:	x^*	= (3.512118414,.2169881741,3.552174034)	
	$f(x^*)$	= 961.7151721	
	$r(x^*)$	= 0	
	$e(x^*)$	= .62E-5	
	μ	= 0	
	$I(x^*)$	= -	
	u^*_{max}/u^*_{min}	= 1.223/.2749 = 4.45	
	$\lambda^*_{max}/\lambda^*_{min}$	= 1.52/1.52 = 1	

PROBLEM:	64
CLASSIFICATION:	PPR-P1-2
SOURCE:	Best [7]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 3$

OBJECTIVE FUNCTION:

$$f(x) = 5x_1 + 50000/x_1 + 20x_2 + 72000/x_2 + 10x_3 + 144000/x_3$$

CONSTRAINTS:

$$1 - 4/x_1 - 32/x_2 - 120/x_3 \geq 0$$

$$1.E-5 \leq x_i \quad , \quad i=1,2,3$$

START:	x_o	= (1 , 1 , 1)	(not feasible)
	$f(x_o)$	= 266035	
SOLUTION:	x^*	= (108.7347175,85.12613942,204.3247078)	
	$f(x^*)$	= 6299.842428	
	$r(x^*)$	= 0	
	$e(x^*)$	= .28E-4	
	μ	= 1	
	$I(x^*)$	= (1)	
	u^*_{max}/u^*_{min}	= 2279/2279 = 1	
	$\lambda^*_{max}/\lambda^*_{min}$	= .21/.092 = 2.28	

PROBLEM:	65
CLASSIFICATION:	QQR-P1-3
SOURCE:	Murtagh, Sargent [47]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - x_2)^2 + (x_1 + x_2 - 10)^2/9 + (x_3 - 5)^2$$

CONSTRAINTS:

$$48 - x_1^2 - x_2^2 - x_3^2 \geq 0$$

$$-4.5 \leq x_i \leq 4.5 \quad , \quad i=1,2$$

$$-5 \leq x_3 \leq 5$$

START:	x_o	=	(-5 , 5 , 0) (not feasible)
	$f(x_o)$	=	1225/9
SOLUTION:	x^*	=	(3.650461821,3.65046168,4.6204170507)
	$f(x^*)$	=	.9535288567
	$r(x^*)$	=	0
	$e(x^*)$	=	.40E-6
	μ	=	1
	$I(x^*)$	=	(1)
	u^*_{max}/u^*_{min}	=	.08215/.08215 = 1
	$\lambda^*_{max}/\lambda^*_{min}$	=	1.95/1.68 = 1.16

PROBLEM:	66
CLASSIFICATION:	LGR-P1-1
SOURCE:	Eckhardt [24]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = .2x_3 - .8x_1$$

CONSTRAINTS:

$$x_2 - \exp(x_1) \geq 0$$

$$x_3 - \exp(x_2) \geq 0$$

$$0 \leq x_1 \leq 100$$

$$0 \leq x_2 \leq 100$$

$$0 \leq x_3 \leq 10$$

START:	x_o	=	(0 , 1.05 , 2.9) (feasible)
	$f(x_o)$	=	.58
SOLUTION:	x^*	=	(.1841264879,1.202167873,3.327322322)
	$f(x^*)$	=	.5181632741
	$r(x^*)$	=	.58E-10
	$e(x^*)$	=	.86E-11
	μ	=	2
	$I(x^*)$	=	(1 , 2)
	u^*_{max}/u^*_{min}	=	.6654/.2 = 3.33
	$\lambda^*_{max}/\lambda^*_{min}$	=	.096/.096 = 1

PROBLEM:	67 (Colville No.8)
CLASSIFICATION:	GGI-P1-1
SOURCE:	Colville [20], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 3$
NUMBER OF CONSTRAINTS:	$m_1 = 14$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = -(.063y_2(x)y_5(x) - 5.04x_1 - 3.36y_3(x) - .035x_2 - 10x_3)$$

$y_i(x)$: cf. Appendix A

CONSTRAINTS:

$$y_{i+1}(x) - a_i \geq 0 \quad , \quad i=1,\ldots,7$$

$$a_i - y_{i-6}(x) \geq 0 \quad , \quad i=8,\ldots,14$$

$$1.E-5 \leq x_1 \leq 2.E3$$

$$1.E-5 \leq x_2 \leq 1.6E4$$

$$1.E-5 \leq x_3 \leq 1.2E2$$

a_i : cf. Appendix A

START:	x_o	= (1745 , 12000 , 110)	(feasible)
	$f(x_o)$	= 868.6458	
SOLUTION:	x^*	= (1728.371286,16000.00000,98.14151402)	
	$f(x^*)$	= -1162.036507	
	$r(x^*)$	= 0	
	$e(x^*)$	= 0	
	μ	= 3	
	$I(x^*)$	= (9 , 11 , 19)	
	u^*_{max}/u^*_{min}	= 1.5872/.03403 = 46.6	
	$\lambda^*_{max}/\lambda^*_{min}$	= -	

PROBLEM:	68 , 69 (cost optimal inspection plan)
CLASSIFICATION:	GGR-P1-(1,2)
SOURCE:	Collani [19]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = \left(a_i n_i - \frac{b_i(\exp(x_1) - 1) - x_3}{\exp(x_1) - 1 + x_4} x_4\right)/x_1 \quad , \quad i=1,2$$

No. 68 : $a_1 = .0001$, $b_1 = 1$, $d_1 = 1$, $n_1 = 24$

No. 69 : $a_2 = .1$, $b_2 = 1000$, $d_2 = 1$, $n_2 = 4$

CONSTRAINTS:

$$x_3 - 2\Phi(-x_2) = 0$$

$$x_4 - \Phi(-x_2 + d_i\sqrt{n_i}) - \Phi(-x_2 - d_i\sqrt{n_i}) = 0$$

$$\Phi(x) = \int_{-\infty}^{x} \exp(-y^2/2)/\sqrt{2\pi} \; dy$$

$$.0001 \le x_1 \le 100 \;,\; 0 \le x_2 \le 100 \;,\; 0 \le x_3 \le 2 \;,\; 0 \le x_4 \le 2$$

START:	x_o	=	(1 , 1 , 1 , 1) (not feasible) - for both problems	
	$f(x_o)$	=	-.2618407	-631.3525
SOLUTION:	x^*	=	(.06785874,3.6461717,.00026617,.8948622)	
			(.02937141,1.1902534,.23394676,.7916678)	
	$f(x^*)$	=	-.920425026	-956.71288
	$r(x^*)$	=	.54E-7	.44E-10
	$e(x^*)$	=	.14E-4	.33E-4
	μ	=	0	0
	$I(x^*)$	=	-	-
	u^*_{max}/u^*_{min}	=	13.66/.0777 = 176	44.47/32.81 = 1.3
	$\lambda^*_{max}/\lambda^*_{min}$	=	16.4/.062 = .26E3	.26E5/19.6 = .1E4

PROBLEM:	70
CLASSIFICATION:	SQR-P1-1
SOURCE:	Himmelblau [29,30]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{i=1}^{19} (y_{i,cal} - y_{i,obs})^2$$

$$y_{i,cal} = (1 + \frac{1}{12x_2})[x_3 b^{x_2} (x_2/6.2832)^{.5} (c_i/7.685)^{x_2-1} \exp(x_2 - bc_i x_2/7.658)] + (1 + \frac{1}{12x_1})[(1 - x_3)(b/x_4)^{x_1} (x_1/6.2832)^{.5} (c_i/7.658)^{x_1-1} \exp(x_1 - bc_i x_1/(7.658x_4))]$$

$$b = x_3 + (1 - x_3)x_4$$

c_i, $y_{i,obs}$: cf. Appendix A

CONSTRAINTS:

$$x_3 + (1 - x_3)x_4 \geq 0$$

$$.00001 \leq x_i \leq 100 \quad , \quad i=1,2,4$$

$$.00001 \leq x_3 \leq 1$$

START:	x_o	=	(2 , 4 , .04 , 2) (feasible)
	$f(x_o)$	=	.9818596
SOLUTION:	x^*	=	(12.27695,4.631788,.3128625,2.029290)
	$f(x^*)$	=	.007498464
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	-
	$\lambda^*_{max}/\lambda^*_{min}$	=	18.1/.91E-3 = .20E5

PROBLEM:	71
CLASSIFICATION:	PPR-P1-3
SOURCE:	Bartholomew-Biggs [4]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 1$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = x_1x_4(x_1 + x_2 + x_3) + x_3$$

CONSTRAINTS:

$$x_1x_2x_3x_4 - 25 \geq 0$$

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 - 40 = 0$$

$$1 \leq x_i \leq 5 \quad , \quad i=1,\ldots,4$$

START:	x_o	=	(1 , 5 , 5 , 1) (feasible)
	$f(x_o)$	=	16
SOLUTION:	x^*	=	(1 , 4.7429994 , 3.8211503 , 1.3794082)
	$f(x^*)$	=	17.0140173
	$r(x^*)$	=	0
	$e(x^*)$	=	.51E-6
	μ	=	2
	$I(x^*)$	=	(1 , 2)
	u^*_{max}/u^*_{min}	=	1.0879/.1615 = 6.74
	$\lambda^*_{max}/\lambda^*_{min}$	=	1.18/1.18 = 1

PROBLEM:	72 (optimal sample size)
CLASSIFICATION:	LPR-P1-1
SOURCE:	Bracken, McCormick [13]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = 1 + x_1 + x_2 + x_3 + x_4$$

CONSTRAINTS:

$$.0401 - 4/x_1 - 2.25/x_2 - 1/x_3 - .25/x_4 \geq 0$$

$$.010085 - .16/x_1 - .36/x_2 - .64/x_3 - .64/x_4 \geq 0$$

$$.001 \leq x_i \leq (5 - i)E5 \quad , \quad i=1,\ldots,4$$

START:	x_o	= (1 , 1 , 1 , 1)	(not feasible)
	$f(x_o)$	= 5	
SOLUTION:	x^*	= (193.4071,179.5475,185.0186,168.7062)	
	$f(x^*)$	= 727.67937	
	$r(x^*)$	= 0	
	$e(x^*)$	= .11E-4	
	μ	= 2	
	$I(x^*)$	= (1 , 2)	
	u^*_{max}/u^*_{min}	= .4147E5/.7693E4 = 5.39	
	$\lambda^*_{max}/\lambda^*_{min}$	= .011/.011 = 1.02	

PROBLEM:	73 (cattle-feed)
CLASSIFICATION:	LGI-P1-1
SOURCE:	Biggs [10], Bracken, McCormick [13]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 2(1)$, $m-m_1 = 1(1)$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = 24.55x_1 + 26.75x_2 + 39x_3 + 40.50x_4$$

CONSTRAINTS:

$$2.3x_1 + 5.6x_2 + 11.1x_3 + 1.3x_4 - 5 \geq 0$$

$$12x_1 + 11.9x_2 + 41.8x_3 + 52.1x_4 - 21$$
$$- 1.645(.28x_1^2 + .19x_2^2 + 20.5x_3^2 + .62x_4^2)^{1/2} \geq 0$$

$$x_1 + x_2 + x_3 + x_4 - 1 = 0$$

$$0 \leq x_i \quad , \quad i=1,\ldots,4$$

START:	x_o	=	(1 , 1 , 1 , 1) (not feasible)
	$f(x_o)$	=	130.8
SOLUTION:	x^*	=	(.6355216,-.12E-11,.3127019,.05177655)
	$f(x^*)$	=	29.894378
	$r(x^*)$	=	.99E-10
	$e(x^*)$	=	0
	μ	=	3
	$I(x^*)$	=	(1 , 2 , 4)
	u^*_{max}/u^*_{min}	=	18.37/.2433 = 75.5
	$\lambda^*_{max}/\lambda^*_{min}$	=	-

PROBLEM:	74 , 75
CLASSIFICATION:	PGR-P1-(1,2)
SOURCE:	Beuneu [9]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 2(2)$, $m-m_1 = 3$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = 3x_1 + 1.E\text{-}6x_1^3 + 2x_2 + \tfrac{2}{3}E\text{-}6x_2^3$$

CONSTRAINTS:

$$x_4 - x_3 + a_j \geq 0$$

$$x_3 - x_4 + a_j \geq 0$$

$$1000\sin(-x_3 - .25) + 1000\sin(-x_4 - .25) + 894.8 - x_1 = 0$$

$$1000\sin(x_3 - .25) + 1000\sin(x_3 - x_4 - .25) + 894.8 - x_2 = 0$$

$$1000\sin(x_4 - .25) + 1000\sin(x_4 - x_3 - .25) + 1294.8 = 0$$

$$0 \leq x_i \leq 1200 \quad , \quad i=1,2$$

$$-a_j \leq x_i \leq a_j \quad , \quad i=3,4$$

No. 74 : $a_1 = .55$ No.75 : $a_2 = .48$

START:	x_o	=	(0 , 0 , 0 , 0) - for both problems	(not feasible)
	$f(x_o)$	=	0	
SOLUTION:	x^*	=	(679.9453,1026.067,.1188764,-.3962336)	
			(776.1592,925.1949,.05110879, -.4288911)	
	$f(x^*)$	=	5126.4981	5174.4129
	$r(x^*)$	=	.75E-7	.30E-7
	$e(x^*)$	=	.52E-7	0
	μ	=	0	1
	$I(x^*)$	=	-	(1)
	u^*_{max}/u^*_{min}	=	5.46/4.11 = 1.33	2779/3.712 = 748.7
	$\lambda^*_{max}/\lambda^*_{min}$	=	.49E-2/.49E-2 = 1	-

PROBLEM:	76
CLASSIFICATION:	QLR-P1-1
SOURCE:	Murtagh, Sargent [47]
NUMBER OF VARIABLES:	$n = 4$
NUMBER OF CONSTRAINTS:	$m_1 = 3(3)$, $m-m_1 = 0$, $b = 4$

OBJECTIVE FUNCTION:

$$f(x) = x_1^2 + .5x_2^2 + x_3^2 + .5x_4^2 - x_1x_3 + x_3x_4 - x_1 - 3x_2 + x_3 - x_4$$

CONSTRAINTS:

$$5 - x_1 - 2x_2 - x_3 - x_4 \geq 0$$

$$4 - 3x_1 - x_2 - 2x_3 + x_4 \geq 0$$

$$x_2 + 4x_3 - 1.5 \geq 0$$

$$0 \leq x_i \quad , \quad i=1,\ldots,4$$

START:	x_o	= (.5 , .5 , .5 , .5)	(feasible)
	$f(x_o)$	= -1.25	
SOLUTION:	x^*	= (.2727273,2.090909,-.26E-10,.5454545)	
	$f(x^*)$	= -4.681818181	
	$r(x^*)$	= .84E-10	
	$e(x^*)$	= .15E-10	
	μ	= 2	
	$I(x^*)$	= (1 , 6)	
	u^*_{max}/u^*_{min}	= 1.7272/.4545 = 3.8	
	$\lambda^*_{max}/\lambda^*_{min}$	= 1.83/1 =1.83	

PROBLEM:	77
CLASSIFICATION:	PGR-P1-3
SOURCE:	Betts [8], Miele e.al. [42,44,45]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 1)^2 + (x_1 - x_2)^2 + (x_3 - 1)^2 + (x_4 - 1)^4 + (x_5 - 1)^6$$

CONSTRAINTS:

$$x_1^2 x_4 + \sin(x_4 - x_5) - 2\sqrt{2} = 0$$

$$x_2 + x_3^4 x_4^2 - 8 - \sqrt{2} = 0$$

START:	x_o	=	(2 , 2 , 2 , 2 , 2 , 2)
	$f(x_o)$	=	4 (not feasible)
SOLUTION:	x^*	=	(1.166172,1.182111,1.380257,1.506036,
	$f(x^*)$	=	.24150513 .6109203)
	$r(x^*)$	=	.12E-9
	$e(x^*)$	=	.53E-7
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	.08554/.03188 = 2.68
	$\lambda^*_{max}/\lambda^*_{min}$	=	3.92/.75 = 5.25

PROBLEM:	78
CLASSIFICATION:	PPR-P1-4
SOURCE:	Asaadi [1], Powell [51]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = x_1x_2x_3x_4x_5$$

CONSTRAINTS:

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 - 10 = 0$$

$$x_2x_3 - 5x_4x_5 = 0$$

$$x_1^3 + x_2^3 + 1 = 0$$

START:	x_o	=	(-2 , 1.5 , 2 , -1 , -1)
	$f(x_o)$	=	-6 (not feasible)
SOLUTION:	x^*	=	(-1.717142,1.595708,1.827248,-.7636429, -.7636435)
	$f(x^*)$	=	-2.91970041
	$r(x^*)$	=	.35E-9
	$e(x^*)$	=	.91E-5
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	.7444/.09681 = 7.69
	$\lambda^*_{max}/\lambda^*_{min}$	=	3.04/2.98 = 1.02

PROBLEM:	79
CLASSIFICATION:	PPR-P1-5
SOURCE:	Betts [8], Miele e.al. [42,44,45]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 1)^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_3 - x_4)^4 + (x_4 - x_5)^4$$

CONSTRAINTS:

$$x_1 + x_2^2 + x_3^3 - 2 - 3\sqrt{2} = 0$$

$$x_2 - x_3^2 + x_4 + 2 - 2\sqrt{2} = 0$$

$$x_1 x_5 - 2 = 0$$

START:	x_o	=	(2 , 2 , 2 , 2 , 2) (not feasible)
	$f(x_o)$	=	1
SOLUTION:	x^*	=	(1.191127,1.362603,1.472818,1.635017,
	$f(x^*)$	=	.0787768209 1.679081)
	$r(x^*)$	=	.58E-9
	$e(x^*)$	=	.71E-10
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	.3882E-1/.2873E-3 = 135.1
	$\lambda^*_{max}/\lambda^*_{min}$	=	2.03/.70 = 2.88

PROBLEM:	80
CLASSIFICATION:	GPR-P1-1
SOURCE:	Powell [52]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3$, $b = 10$

OBJECTIVE FUNCTION:

$$f(x) = \exp(x_1 x_2 x_3 x_4 x_5)$$

CONSTRAINTS:

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 - 10 = 0$$

$$x_2 x_3 - 5 x_4 x_5 = 0$$

$$x_1^3 + x_2^3 + 1 = 0$$

$$-2.3 \le x_i \le 2.3 \quad , \quad i=1,2$$

$$-3.2 \le x_i \le 3.2 \quad , \quad i=3,4,5$$

START:	x_o	=	(-2 , 2 , 2 , -1 , -1) (not feasible)
	$f(x_o)$	=	3.3546E-4
SOLUTION:	x^*	=	(-1.717143,1.595709,1.827247,-.7636413,
	$f(x^*)$	=	.0539498478 -.7636450)
	$r(x^*)$	=	.41E-9
	$e(x^*)$	=	.49E-6
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	.04016/.005222 = 7.69
	$\lambda^*_{max}/\lambda^*_{min}$	=	.16/.16 = 1.02

PROBLEM:	81
CLASSIFICATION:	GPR-P1-2
SOURCE:	Powell [52]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3$, $b = 10$

OBJECTIVE FUNCTION:

$$f(x) = \exp(x_1x_2x_3x_4x_5) - .5(x_1^3 + x_2^3 + 1)^2$$

CONSTRAINTS:

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 - 10 = 0$$

$$x_2x_3 - 5x_4x_5 = 0$$

$$x_1^3 + x_2^3 + 1 = 0$$

$$-2.3 \le x_i \le 2.3 \quad , \quad i=1,2$$

$$-3.2 \le x_i \le 3.2 \quad , \quad i=3,4,5$$

START:	x_o	=	(-2 , 2 , 2 , -1 , -1) (not feasible)
	$f(x_o)$	=	-.49966
SOLUTION:	x^*	=	(-1.717142,1.159571,1.827248,-.7636474,
	$f(x^*)$	=	.0539498478 -.7636390)
	$r(x^*)$	=	.21E-9
	$e(x^*)$	=	.11E-5
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	.04016/.005223 = 7.69
	$\lambda^*_{max}/\lambda^*_{min}$	=	.16/.16 = 1.02

PROBLEM:	83 (Colville No.3)
CLASSIFICATION:	QQR-P1-4
SOURCE:	Colville [20], Dembo [22], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 6$, $m-m_1 = 0$, $b = 10$

OBJECTIVE FUNCTION:

$$f(x) = 5.3578547x_3^2 + .8356891x_1x_5 + 37.293239x_1 - 40792.141$$

CONSTRAINTS:

$$92 \geq a_1 + a_2x_2x_5 + a_3x_1x_4 - a_4x_3x_5 \geq 0$$

$$20 \geq a_5 + a_6x_2x_5 + a_7x_1x_2 + a_8x_3^2 - 90 \geq 0$$

$$5 \geq a_9 + a_{10}x_3x_5 + a_{11}x_1x_3 + a_{12}x_3x_4 - 20 \geq 0$$

$$78 \leq x_1 \leq 102$$

$$33 \leq x_2 \leq 45$$

$$27 \leq x_i \leq 45 \quad , \quad i=3,4,5$$

a_i : cf. Appendix A

START:	x_o	=	(78 , 33 , 27 , 27 , 27) (not feasible)
	$f(x_o)$	=	-32217
SOLUTION:	x^*	=	(78,33,29.99526,45,36.77581)
	$f(x^*)$	=	-30665.53867
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	5
	$I(x^*)$	=	(3 , 4 , 7 , 8 , 15)
	u^*_{max}/u^*_{min}	=	809.4/26.64 = 30.4
	$\lambda^*_{max}/\lambda^*_{min}$	=	-

PROBLEM:	84
CLASSIFICATION:	QQR-P1-5
SOURCE:	Betts [8], Box [11,12], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 6$, $m-m_1 = 0$, $b = 10$

OBJECTIVE FUNCTION:

$$f(x) = -a_1 - a_2x_1 - a_3x_1x_2 - a_4x_1x_3 - a_5x_1x_4 - a_6x_1x_5$$

CONSTRAINTS:

$$294000 \geq a_7x_1 + a_8x_1x_2 + a_9x_1x_3 + a_{10}x_1x_4 + a_{11}x_1x_5 \geq 0$$

$$294000 \geq a_{12}x_1 + a_{13}x_1x_2 + a_{14}x_1x_3 + a_{15}x_1x_4 + a_{16}x_1x_5 \geq 0$$

$$277200 \geq a_{17}x_1 + a_{18}x_1x_2 + a_{19}x_1x_3 + a_{20}x_1x_4 + a_{21}x_1x_5 \geq 0$$

$$0 \leq x_1 \leq 1000$$

$$1.2 \leq x_2 \leq 2.4$$

$$20 \leq x_3 \leq 60$$

$$9 \leq x_4 \leq 9.3$$

$$6.5 \leq x_5 \leq 7$$

a_i : cf. Appendix A

START:	x_o	=	(2.52 , 2 , 37.5 , 9.25 , 6.8)
	$f(x_o)$	=	-2351243.5 (feasible)
SOLUTION:	x^*	=	(4.53743097 , 2.4 , 60 , 9.3 , 7)
	$f(x^*)$	=	-5280335.133
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	5
	$I(x^*)$	=	(6 , 13 , 14 , 15 , 16)
	u^*_{max}/u^*_{min}	=	.7168E6/.1914E2 = .37E5
	$\lambda^*_{max}/\lambda^*_{min}$	=	-

PROBLEM:	85
CLASSIFICATION:	GGI-P1-2
SOURCE:	Barness [2], Caroll [17], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 38(3)$, $m-m_1 = 0$, $b = 10$

OBJECTIVE FUNCTION:

$$f(x) = -5.843E\text{-}7y_{17}(x) + 1.17E\text{-}4y_{14}(x) + 2.358E\text{-}5y_{13}(x) + 1.502E\text{-}6y_{16}(x) + .0321y_{12}(x) + .00423y_5(x) + 1.E\text{-}4c_{15}(x)/c_{16}(x) + 37.48y_2(x)/c_{12}(x) - .1365$$

CONSTRAINTS:

$1.5x_2 - x_3 \geq 0$

$y_1(x) - 213.1 \geq 0$

$405.23 - y_1(x) \geq 0$

$y_{j-2}(x) - a_{j-2} \geq 0$, $j=4,\ldots,19$

$b_{j-18} - y_{j-18}(x) \geq 0$, $j=20,\ldots,35$

$y_4(x) - .28/.72y_5(x) \geq 0$

$21 - 3496y_2(x)/c_{12}(x) \geq 0$

$62212/c_{17}(x) - 110.6 - y_1(x) \geq 0$

$704.4148 \leq x_1 \leq 906.3855$

$68.6 \leq x_2 \leq 288.88$

$0 \leq x_3 \leq 134.75$

$193 \leq x_4 \leq 287.0966$

$25 \leq x_5 \leq 84.1988$

$y_j(x)$, $c_j(x)$, a_j , b_j : cf. Appendix A

START: $x_o = (900 , 80 , 115 , 267 , 27)$

$f(x_o) = -.939$ (feasible)

SOLUTION: $x^* = (705.1803, 68.60005, 102.90001, 282.324999, 37.5850413)$

$f(x^*) = -1.90513375$

$r(x^*) = 0$

$e(x^*) = 0$

$\mu = 0$

$I(x^*) = -$

$u^*_{max}/u^*_{min} = -$

$\lambda^*_{max}/\lambda^*_{min} = .56E\text{-}3/.46E\text{-}6 = .12E4$

PROBLEM:	86 (Colville No.1)
CLASSIFICATION:	PLR-P1-1
SOURCE:	Colville [20], Himmelblau [29], Murthagh, Sargent [47]
NUMBER OF VARIABLES:	$n = 5$
NUMBER OF CONSTRAINTS:	$m_1 = 10(10)$, $m-m_1 = 0$, $b = 5$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{j=1}^{5} e_j x_j + \sum_{i=1}^{5} \sum_{j=1}^{5} c_{ij} x_i x_j + \sum_{j=1}^{5} d_j x_j^3$$

CONSTRAINTS:

$$\sum_{j=1}^{5} a_{ij} x_j - b_i \geq 0 \quad , \quad i=1,\ldots,10$$

$$0 \leq x_i \quad , \quad i=1,\ldots,5$$

$a_{ij}, b_i, c_{ij}, d_j, e_j$: cf. Appendix A

START:	x_o	= (0 , 0 , 0 , 0 , 1)	(feasible)
	$f(x_o)$	= 20	
SOLUTION:	x^*	= (.3,.33346761,.4,.42831010,.22396487)	
	$f(x^*)$	= -32.34867897	
	$r(x^*)$	= .70E-9	
	$e(x^*)$	= .94E-8	
	μ	= 4	
	$I(x^*)$	= (3 , 5 , 6 , 9)	
	u^*_{max}/u^*_{min}	= 11.84/.1039 = 113.9	
	$\lambda^*_{max}/\lambda^*_{min}$	= 68.1/68.1 = 1	

PROBLEM:	87 (Colville No.6)
CLASSIFICATION:	GGI-P1-3
SOURCE:	Colville [20], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 6$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 4$, $b = 12$

OBJECTIVE FUNCTION:

$$f(x) = f_1(x) + f_2(x)$$

$$f_1(x) = \begin{cases} 30x_1 , & 0 \le x_1 < 300 \\ 31x_1 , & 300 \le x_1 \le 400 \end{cases} \qquad f_2(x) = \begin{cases} 28x_2 , & 0 \le x_2 < 100 \\ 29x_2 , & 100 \le x_2 < 200 \\ 30x_2 , & 200 \le x_2 \le 1000 \end{cases}$$

CONSTRAINTS:

$$300 - x_1 - \frac{1}{a} x_3 x_4 \cos(b - x_6) + \frac{c}{a} d x_3^2 = 0 \qquad (a = 131.078)$$

$$- x_2 - \frac{1}{a} x_3 x_4 \cos(b + x_6) + \frac{c}{a} d x_4^2 = 0 \qquad (b = 1.48577)$$

$$- x_5 - \frac{1}{a} x_3 x_4 \sin(b + x_6) + \frac{c}{a} e x_4^2 = 0 \qquad (c = .90798)$$

$$200 - \frac{1}{a} x_3 x_4 \sin(b - x_6) + \frac{c}{a} e x_3^2 = 0 \qquad (d = \cos 1.47588)$$

$$(e = \sin 1.47588)$$

$$0 \le x_1 \le 400 \qquad 340 \le x_3 \le 420 \qquad -1000 \le x_5 \le 10000$$

$$0 \le x_2 \le 1000 \qquad 340 \le x_4 \le 420 \qquad 0 \le x_6 \le .5236$$

START:	x_o	=	(390,1000,419.5,340.5,198.175,.5)
	$f(x_o)$	=	42090 (not feasible)
SOLUTION:	x^*	=	(107.8119,196.3186,373.8307,420, 213.0713,.1532920)
	$f(x^*)$	=	8927.5977
	$r(x^*)$	=	.10E-6
	$e(x^*)$	=	.74E-6
	μ	=	1
	$I(x^*)$	=	(10)
	u^*_{max}/u^*_{min}	=	30/.23E-6 = .13E9
	$\lambda^*_{max}/\lambda^*_{min}$	=	.017/.017 = 1

PROBLEM:	88 - 92 (time-optimal heat conduction)
CLASSIFICATION:	QGR-P1-(1,...,5)
SOURCE:	Schittkowski [54]
NUMBER OF VARIABLES:	$n = 2,\ldots,6$
NUMBER OF CONSTRAINTS:	$m_1 = 1$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{i=1}^{n} x_i^2$$

CONSTRAINTS:

$$\epsilon^2 - h(x) \geq 0$$

$$\epsilon = .01$$

$$h(x) = \int_0^1 \left(\sum_{i=1}^{30} \alpha_j(s)\rho_j(x) - k_o(s)\right)^2 ds$$

$$\alpha_j(s) = \mu_j^2 A_j \cos(\mu_j s)$$

$$\rho_j(x) = -\mu_j^{-2}\left(\exp\left(-\mu_j^2 \sum_{i=1}^{n} x_i^2\right) - 2\exp\left(-\mu_j^2 \sum_{i=2}^{n} x_i^2\right) + \ldots\right.$$

$$\left. + (-1)^{n-1} 2\exp(-\mu_j^2 x_n^2 + (-1)^n)\right)$$

$$k_o(s) = .5(1 - s^2)$$

$$A_j = 2\sin\mu_j/(\mu_j + \sin\mu_j\cos\mu_j) \ , \quad \mu_j : \ \mu \tan\mu = 1$$

START:	x_o	=	$(.5 , -.5 , \ldots , (-1)^{n+1}.5)$
	$f(x_o)$	=	$.25n$ (not feasible)
SOLUTION:	x^*	=	(cf. Appendix A)
	$f(x^*)$	=	1.36265681
	$r(x^*)$	$\leq$	.30E-10 (cf. Appendix A)
	$e(x^*)$	$\leq$	.16E-2 (cf. Appendix A)
	μ	=	1
	$I(x^*)$	=	(1)
	u^*_{max}/u^*_{min}	=	1059.8/1059.8 = 1
	$\lambda^*_{max}/\lambda^*_{min}$	$\leq$	.11E9 (cf. Appendix A)

PROBLEM:	93 (transformer design)
CLASSIFICATION:	PPR-P1-6
SOURCE:	Bartholomew-Biggs [4]
NUMBER OF VARIABLES:	$n = 6$
NUMBER OF CONSTRAINTS:	$m_1 = 2$, $m-m_1 = 0$, $b = 6$

OBJECTIVE FUNCTION:

$$f(x) = .0204x_1x_4(x_1 + x_2 + x_3) + .0187x_2x_3(x_1 + 1.57x_2 + x_4) + .0607x_1x_4x_5^2(x_1 + x_2 + x_3) + .0437x_2x_3x_6^2(x_1 + 1.57x_2 + x_4)$$

CONSTRAINTS:

$$.001x_1x_2x_3x_4x_5x_6 - 2.07 \geq 0$$

$$1 - .00062x_1x_4x_5^2(x_1 + x_2 + x_3) - .00058x_2x_3x_6^2(x_1 + 1.57x_2 + x_4) \geq 0$$

$$0 \leq x_i \quad , \quad i=1,\ldots,6$$

START:	x_o	=	(5.54,4.4,12.02,11.82,.702,.852)
	$f(x_o)$	=	137.066 (feasible)
SOLUTION:	x^*	=	(5.332666,4.656744,10.43299,12.08230, .7526074,.87865084)
	$f(x^*)$	=	135.075961
	$r(x^*)$	=	.77E-7
	$e(x^*)$	=	.77E-6
	μ	=	2
	$I(x^*)$	=	(1 , 2)
	u^*_{max}/u^*_{min}	=	71.46/62.15 = 1.15
	$\lambda^*_{max}/\lambda^*_{min}$	=	118.9/.21 = 562.9

PROBLEM:	95 - 98
CLASSIFICATION:	LQR-P1-(1,...,4)
SOURCE:	Himmelblau [29], Holzman [32]
NUMBER OF VARIABLES:	$n = 6$
NUMBER OF CONSTRAINTS:	$m_1 = 4$, $m-m_1 = 0$, $b = 12$

OBJECTIVE FUNCTION:

$$f(x) = 4.3x_1 + 31.8x_2 + 63.3x_3 + 15.8x_4 + 68.5x_5 + 4.7x_6$$

CONSTRAINTS:

$$17.1x_1 + 38.2x_2 + 204.2x_3 + 212.3x_4 + 623.4x_5 + 1495.5x_6$$
$$- 169x_1x_3 - 3580x_3x_5 - 3810x_4x_5 - 18500x_4x_6 - 24300x_5x_6 \geq b_1$$

$$17.9x_1 + 36.8x_2 + 113.9x_3 + 169.7x_4 + 337.8x_5 + 1385.2x_6$$
$$- 139x_1x_3 - 2450x_4x_5 - 16600x_4x_6 - 17200x_5x_6 \geq b_2$$

$$-273x_2 - 70x_4 - 819x_5 + 26000x_4x_5 \geq b_3$$

$$159.9x_1 - 311x_2 + 587x_4 + 391x_5 + 2198x_6 - 14000x_1x_6 \geq b_4$$

$$0 \leq x_1 \leq .31 \;,\; 0 \leq x_3 \leq .068 \;,\; 0 \leq x_5 \leq .028$$
$$0 \leq x_2 \leq .046 \;,\; 0 \leq x_4 \leq .042 \;,\; 0 \leq x_6 \leq .0134$$

4 different data vectors b : cf. Appendix A

START:	x_o	= (0 , 0 , 0 , 0 , 0 , 0)	(not feasible)	
	$f(x_o)$	= 0		
SOLUTION:	x^*	= (0,0,0,0,0,.0033233033)	(95,96)	
		= (.2685649,0,0,0,.028,.0134)	(97,98)	
	$f(x^*)$	= .015619514 (95,96)	3.1358091 (97,98)	
	$r(x^*)$	= .21E-9 (95,96)	0	
	$e(x^*)$	= 0		
	μ	= 6		
	$I(x^*)$	= (1,5,6,7,8,9) (95,96)	(1,6,7,8,15,16)	
	u^*_{max}/u^*_{min}	= 66.8/.003=.2E5(95,96)	200/.251=.8E3	
	$\lambda^*_{max}/\lambda^*_{min}$	= -		

PROBLEM:	99
CLASSIFICATION:	GGR-P1-3
SOURCE:	Betts [8]
NUMBER OF VARIABLES:	$n = 7$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 2$, $b = 14$

OBJECTIVE FUNCTION:

$$f(x) = -r_8(x)^2$$

$$r_1(x) = 0 \ , \quad r_i(x) = a_i(t_i - t_{i-1})\cos x_{i-1} + r_{i-1}(x), \ i=2,..,8$$

CONSTRAINTS:

$$q_8(x) - 1.E5 = 0$$

$$s_8(x) - 1.E3 = 0$$

$$0 \le x_i \le 1.58 \ , \ i=1,...,7$$

$$q_1(x) = s_1(x) = 0$$

$$q_i(x) = .5(t_i - t_{i-1})^2(a_i \sin x_{i-1} - b) + (t_i - t_{i-1})s_{i-1}(x) + q_{i-1}(x)$$

$$s_i(x) = (t_i - t_{i-1})(a_i \sin x_{i-1} - b) + s_{i-1}(x) \ , \quad i=2,...,8$$

a_i, t_i, b : cf. Appendix A

START:	x_o	=	(.5 , .5 , .5 , .5 , .5 , .5 , .5)
	$f(x_o)$	=	-.7763605E9 (not feasible)
SOLUTION:	x^*	=	(.5424603,.5290159,.5084506,.4802693, .4512352,.4091878,.3527847)
	$f(x^*)$	=	-.831079892E9
	$r(x^*)$	=	.30E-7
	$e(x^*)$	=	.31E4 , $\|\nabla f(x^*)\|$ = .32E9
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	.1934E5/.4194E2 = .46E3
	$\lambda^*_{max}/\lambda^*_{min}$	=	.50E9/.84E8 = 5.95

PROBLEM:	100
CLASSIFICATION:	PPR-P1-7
SOURCE:	Asaadi [1], Charalambous [18], Wong [59]
NUMBER OF VARIABLES:	$n = 7$
NUMBER OF CONSTRAINTS:	$m_1 = 4$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = (x_1 - 10)^2 + 5(x_2 - 12)^2 + x_3^4 + 3(x_4 - 11)^2 + 10x_5^6 + 7x_6^2 + x_7^4 - 4x_6x_7 - 10x_6 - 8x_7$$

CONSTRAINTS:

$$127 - 2x_1^2 - 3x_2^4 - x_3 - 4x_4^2 - 5x_5 \geq 0$$

$$282 - 7x_1 - 3x_2 - 10x_3^2 - x_4 + x_5 \geq 0$$

$$196 - 23x_1 - x_2^2 - 6x_6^2 + 8x_7 \geq 0$$

$$-4x_1^2 - x_2^2 + 3x_1x_2 - 2x_3^2 - 5x_6 + 11x_7 \geq 0$$

START:	x_o	=	(1 , 2 , 0 , 4 , 0 , 1 , 1)
	$f(x_o)$	=	714 (feasible)
SOLUTION:	x^*	=	(2.330499,1.951372,-.4775414,4.365726, -.6244870,1.038131,1.594227)
	$f(x^*)$	=	680.6300573
	$r(x^*)$	=	.90E-7
	$e(x^*)$	=	.36E-8
	μ	=	2
	$I(x^*)$	=	(1 , 4)
	u^*_{max}/u^*_{min}	=	1.140/.3686 = 3.09
	$\lambda^*_{max}/\lambda^*_{min}$	=	46.6/4.34 = 10.7

PROBLEM:	101 - 103
CLASSIFICATION:	PPR-P1-(8,9,10)
SOURCE:	Beck, Ecker [5], Dembo [22]
NUMBER OF VARIABLES:	$n = 7$
NUMBER OF CONSTRAINTS:	$m_1 = 6$, $m-m_1 = 0$, $b = 14$

OBJECTIVE FUNCTION: 101 : a=-.25 , 102 : a=.125 , 103 : a=.5

$$f(x) = 10x_1x_2^{-1}x_4^{2}x_6^{-3}x_7^{a} + 15x_1^{-1}x_2^{-2}x_3x_4x_5^{-1}x_7^{-.5} + 20x_1^{-2}x_2x_4^{-1}x_5^{-2}x_6 + 25x_1^{2}x_2^{2}x_3^{-1}x_5^{.5}x_6^{-2}x_7$$

CONSTRAINTS:

$$1 - .5x_1^{.5}x_3^{-1}x_6^{-2}x_7 - .7x_1^{3}x_2x_3^{-2}x_6x_7^{.5} - .2x_2^{-1}x_3x_4^{-.5}x_6^{2/3}x_7^{1/4} \geq 0$$

$$1 - 1.3x_1^{-.5}x_2x_3^{-1}x_5^{-1}x_6 - .8x_3x_4^{-1}x_5^{-1}x_6^{2} - 3.1x_1^{-1}x_2^{.5}x_4^{-2}x_5^{-1}x_6^{1/3} \geq 0$$

$$1 - 2x_1x_3^{-1.5}x_5x_6^{-1}x_7^{1/3} - .1x_2x_3^{-.5}x_5x_6^{-1}x_7^{-.5} - x_1^{-1}x_2x_3^{.5}x_5 - .65x_2^{-2}x_3x_5x_6^{-1}x_7 \geq 0$$

$$1 - .2x_1^{-2}x_2x_4^{-1}x_5^{.5}x_7^{1/3} - .3x_1^{.5}x_2^{2}x_3x_4^{1/3}x_7^{1/4}x_5^{-2/3} - .4x_1^{-3}x_2^{-2}x_3x_5x_7^{3/4} - .5x_3^{-2}x_4x_7^{.5} \geq 0$$

$$100 \leq f(x) \leq 3000 \ , \ .1 \leq x_i \leq 10 \ , \ i=1,..,6, \ .01 \leq x_7 \leq 10$$

START: x_o = (6 , 6 , 6 , 6 , 6 , 6 , 6) (not feas.)

$f(x_o)$ = 2205.868 , 2206.889 , 2208.886

SOLUTION: x^* = (cf. Appendix A)

		101	102	103
$f(x^*)$	=	1809.76476	911.880571	543.667958
$r(x^*)$	=	.10E-10	.27E-10	.61E-11
$e(x^*)$	=	.59E-6	.14E-6	.92E-7
μ	=	3	3	4
$I(x^*)$	=	(2,3,13)	(1,2,3)	(1,2,3,4)
u^*_{max}/u^*_{min}	=	4567/.81E-6	2173/21.13	1287/38.71
$\lambda^*_{max}/\lambda^*_{min}$	=	.43E4/.51E2=85	.23E4/.32E2=73	.46E3/.88E2=5.2

PROBLEM:	104 (optimal reactor design)
CLASSIFICATION:	PPR-P1-11
SOURCE:	Dembo [22], Rijckaert [53]
NUMBER OF VARIABLES:	$n = 8$
NUMBER OF CONSTRAINTS:	$m_1 = 6$, $m-m_1 = 0$, $b = 16$

OBJECTIVE FUNCTION:

$$f(x) = .4x_1^{.67}x_7^{-.67} + .4x_2^{.67}x_8^{-.67} + 10 - x_1 - x_2$$

CONSTRAINTS:

$$1 - .0588x_5x_7 - .1x_1 \geq 0$$

$$1 - .0588x_6x_8 - .1x_1 - .1x_2 \geq 0$$

$$1 - 4x_3x_5^{-1} - 2x_3^{-.71}x_5^{-1} - .0588x_3^{-1.3}x_7 \geq 0$$

$$1 - 4x_4x_6^{-1} - 2x_4^{-.71}x_6^{-1} - .0588x_4^{-1.3}x_8 \geq 0$$

$$1 \leq f(x) \leq 4.2$$

$$.1 \leq x_i \leq 10 \quad , \quad i=1,\ldots,8$$

START:	x_o	=	(6,3,.4,.2,6,6,1,.5)	(not feasible)
	$f(x_o)$	=	3.65	
SOLUTION:	x^*	=	(6.465114,2.232709,.6673975,.5957564, 5.932676,5.527235,1.013322,.4006682)	
	$f(x^*)$	=	3.9511634396	
	$r(x^*)$	=	.58E-10	
	$e(x^*)$	=	.31E-10	
	μ	=	4	
	$I(x^*)$	=	(1 , 2 , 3 , 4)	
	u^*_{max}/u^*_{min}	=	6.206/.8472 = 7.32	
	$\lambda^*_{max}/\lambda^*_{min}$	=	1.87/.043 = 43.2	

PROBLEM:	105 (maximum-likelihood estimation)
CLASSIFICATION:	GLR-P1-2
SOURCE:	Bracken, McCormick [13]
NUMBER OF VARIABLES:	$n = 8$
NUMBER OF CONSTRAINTS:	$m_1 = 1(1)$, $m-m_1 = 0$, $b = 16$

OBJECTIVE FUNCTION:

$$f(x) = -\sum_{i=1}^{235} \ln((a_i(x) + b_i(x) + c_i(x))/\sqrt{2\pi})$$

$$a_i(x) = x_1/x_6 \exp(-(y_i - x_3)^2/(2x_6^2))$$

$$b_i(x) = x_2/x_7 \exp(-(y_i - x_4)^2/(2x_7^2)) \quad , \quad i=1,\ldots,235$$

$$c_i(x) = (1 - x_2 - x_1)/x_8 \exp(-(y_i - x_5)^2/(2x_8^2))$$

y_i : cf. Appendix A

CONSTRAINTS:

$1 - x_1 - x_2 \geq 0$

$.001 \leq x_i \leq .499$, $i=1,2$ $\quad 100 \leq x_3 \leq 180$

$130 \leq x_4 \leq 210$ $\quad 170 \leq x_5 \leq 240$ $\quad 5 \leq x_i \leq 25$, $i=6,7,8$

START:	x_o	=	(.1,.2,100,125,175,11.2,13.2,15.8)
	$f(x_o)$	=	1297.6693 (feasible)
SOLUTION:	x^*	=	(.4128928,.4033526,131.2613,164.3135, 217.4222,12.28018,15.77170,20.74682)
	$f(x^*)$	=	1138.416240
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	-
	$\lambda^*_{max}/\lambda^*_{min}$	=	.26E4/.28E-2 = .92E6

PROBLEM:	106 (heat exchanger design)
CLASSIFICATION:	LQR-P1-5
SOURCE:	Avriel, Williams [2], Dembo [22]
NUMBER OF VARIABLES:	$n = 8$
NUMBER OF CONSTRAINTS:	$m_1 = 6(3)$, $m-m_1 = 0$, $b = 16$

OBJECTIVE FUNCTION:

$$f(x) = x_1 + x_2 + x_3$$

CONSTRAINTS:

$$1 - .0025(x_4 + x_6) \geq 0$$

$$1 - .0025(x_5 + x_7 - x_4) \geq 0$$

$$1 - .01(x_8 - x_5) \geq 0$$

$$x_1x_6 - 833.33252x_4 - 100x_1 + 83333.333 \geq 0$$

$$x_2x_7 - 1250x_5 - x_2x_4 + 1250x_4 \geq 0$$

$$x_3x_8 - 1250000 - x_3x_5 + 2500x_5 \geq 0$$

$$100 \leq x_1 \leq 10000$$

$$1000 \leq x_i \leq 10000 \quad , \quad i=2,3$$

$$10 \leq x_i \leq 1000 \quad , \quad i=4,\ldots,8$$

START:

x_o = (5000,5000,5000,200,350,150,225,425)

$f(x_o)$ = 15000 (not feasible)

SOLUTION:

x^* = (579.3167,1359.943,5110.071,182,0174, 295.5985,217.9799,286.4162,395,5979)

$f(x^*)$ = 7049.330923

$r(x^*)$ = 0

$e(x^*)$ = .19E-4

μ = 6

$I(x^*)$ = (1 , 2 , 3 , 4 , 5 , 6)

u^*_{max}/u^*_{min} = 5210/.00848 = .61E6

$\lambda^*_{max}/\lambda^*_{min}$ = .81E-3/.38E-3 = 2.11

PROBLEM:	107 (static power scheduling)
CLASSIFICATION:	PGR-P1-4
SOURCE:	Bartholomew-Biggs [4]
NUMBER OF VARIABLES:	$n = 9$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 6$, $b = 8$

OBJECTIVE FUNCTION:

$$f(x) = 3000x_1 + 1000x_1^3 + 2000x_2 + 666.667x_2^3$$

CONSTRAINTS:

$$.4 - x_1 + 2cx_5^2 - x_5x_6(dy_1 + cy_2) - x_5x_7(dy_3 + cy_4) = 0$$

$$.4 - x_2 + 2cx_6^2 + x_5x_6(dy_1 - cy_2) + x_6x_7(dy_5 - cy_6) = 0$$

$$.8 + 2cx_7^2 + x_5x_7(dy_3 - cy_4) - x_6x_7(dy_5 + cy_6) = 0$$

$$.2 - x_3 + 2dx_5^2 + x_5x_6(cy_1 - dy_2) + x_5x_7(cy_3 - dy_4) = 0$$

$$.2 - x_4 + 2dx_6^2 - x_5x_6(cy_1 + dy_2) - x_6x_7(cy_5 + dy_6) = 0$$

$$-.337 + 2dx_7^2 - x_5x_7(cy_3 + dy_4) + x_6x_7(cy_5 - dy_6) = 0$$

$$0 \le x_i \ , \ i=1,2 \quad , \quad .90909 \le x_i \le 1.0909 \ , \ i=5,6,7$$

$$y_1 = \sin x_8 \ , \ y_2 = \cos x_8 \ , \ y_3 = \sin x_9$$

$$y_4 = \cos x_9 \ , \ y_5 = \sin(x_8-x_9) \ , \ y_6 = \cos(x_8-x_9)$$

$$c = (48.4/50.176)\sin.25 \ , \ d = (48.4/50.176)\cos.25$$

START:	x_o	=	(.8,.8,.2,.2,1.0454,1.0454,0,0)
	$f(x_o)$	=	4853.3335 (not feasible)
SOLUTION:	x^*	=	(.6670095,1.022388,.2282879,.1848217, 1.090900,1.090900,1.069036,.1066126, -.3387867)
	$f(x^*)$	=	5055.011803
	$r(x^*)$	=	.18E-9
	$e(x^*)$	=	0
	μ	=	3
	$I(x^*)$	=	(6 , 7 , 8)
	u^*_{max}/u^*_{min}	=	5208/0
	$\lambda^*_{max}/\lambda^*_{min}$	=	-

PROBLEM:	108
CLASSIFICATION:	QQR-P1-6
SOURCE:	Himmelblau [29], Pearson [49]
NUMBER OF VARIABLES:	$n = 9$
NUMBER OF CONSTRAINTS:	$m_1 = 13$, $m-m_1 = 0$, $b = 1$

OBJECTIVE FUNCTION:

$$f(x) = -.5(x_1x_4 - x_2x_3 + x_3x_9 - x_5x_9 + x_5x_8 - x_6x_7)$$

CONSTRAINTS:

$$1 - x_3^2 - x_4^2 \geq 0 \qquad 1 - x_9^2 \geq 0$$

$$1 - x_5^2 - x_6^2 \geq 0 \qquad 1 - x_1^2 - (x_2 - x_9)^2 \geq 0$$

$$1 - (x_1 - x_5)^2 - (x_2 - x_6)^2 \geq 0$$

$$1 - (x_1 - x_7)^2 - (x_2 - x_8)^2 \geq 0$$

$$1 - (x_3 - x_5)^2 - (x_4 - x_6)^2 \geq 0$$

$$1 - (x_3 - x_7)^2 - (x_4 - x_8)^2 \geq 0$$

$$1 - x_7^2 - (x_8 - x_9)^2 \geq 0 \qquad x_1x_4 - x_2x_3 \geq 0$$

$$x_3x_9 \geq 0 \qquad -x_5x_9 \geq 0$$

$$x_5x_8 - x_6x_7 \geq 0 \qquad 0 \leq x_9$$

START: $x_o = (1,1,1,1,1,1,1,1,1)$ (not feasible)

$f(x_o) = 0$

SOLUTION: $x^* = (.8841292,.4672425,.03742076,.9992996,$
$.8841292,.4672424,.03742076,.9992996,$
$.26E-19)$

$f(x^*) = -.8660254038$

$r(x^*) = .39E-9$ $\quad e(x^*) = .33E-11$

$u = 9$ $\quad I(x^*) = (1,3,4,6,7,9,11,12,14)$

$u^*_{max}/u^*_{min} = .1443/0$

$\lambda^*_{max}/\lambda^*_{min} = -$

PROBLEM:	109
CLASSIFICATION:	PGR-P1-5
SOURCE:	Beuneu [9]
NUMBER OF VARIABLES:	$n = 9$
NUMBER OF CONSTRAINTS:	$m_1 = 4(2)$, $m-m_1 = 6$, $b = 16$

OBJECTIVE FUNCTION:

$$f(x) = 3x_1 + 1.E\text{-}6x_1^3 + 2x_2 + .522074E\text{-}6x_2^3$$

CONSTRAINTS:

$$x_4 - x_3 + .55 \geq 0 \qquad x_3 - x_4 + .55 \geq 0$$

$$2250000 - x_1^2 - x_8^2 \geq 0 \qquad 2250000 - x_2^2 - x_9^2 \geq 0$$

$$x_5x_6\sin(-x_3 - \tfrac{1}{4}) + x_5x_7\sin(-x_4 - \tfrac{1}{4}) + 2bx_5^2 - ax_1 + 400a = 0$$

$$x_5x_6\sin(x_3 - \tfrac{1}{4}) + x_6x_7\sin(x_3 - x_4 - \tfrac{1}{4}) + 2bx_6^2 - ax_2 + 400a = 0$$

$$x_5x_7\sin(x_4 - \tfrac{1}{4}) + x_6x_7\sin(x_4 - x_3 - \tfrac{1}{4}) + 2bx_7^2 + 881.779a = 0$$

$$ax_8 + x_5x_6\cos(-x_3 - \tfrac{1}{4}) + x_5x_7\cos(-x_4 - \tfrac{1}{4}) - 200a - 2cx_5^2 + .7533E\text{-}3ax_5^2 = 0$$

$$ax_9 + x_5x_6\cos(x_3 - \tfrac{1}{4}) + x_6x_7\cos(x_3 - x_4 - \tfrac{1}{4}) - 2cx_6^2 + .7533E\text{-}3ax_6^2 - 200a = 0$$

$$x_5x_7\cos(x_4 - \tfrac{1}{4}) + x_6x_7\cos(x_4 - x_3 - \tfrac{1}{4}) - 2cx_7^2 + 22.938a + .7533E\text{-}3ax_7^2 = 0$$

$$0 \leq x_i \ , \ i=1,2 \qquad -.55 \leq x_i \leq .55 \ , \ i=3,4$$

$$196 \leq x_i \leq 252 \ , \ i=5,6,7 \qquad -400 \leq x_i \leq 800 \ , \ i=8,9$$

$$a = 50.176 \ , \quad b = \sin.25 \ , \quad c = \cos.25$$

START: $x_o = (0,\ldots,0)$ $\qquad f(x_o) = 0$ (not feasible)

SOLUTION: $\qquad f(x^*) = 5362.06928$

x^* = (cf. Appendix A)

$r(x^*) = .36E\text{-}7 \qquad e(x^*) = 0$

$\mu = 3 \qquad I(x^*) = (1 , 16 , 17)$

$u^*_{max}/u^*_{min} = 12.53/.13E\text{-}10 = .95E12$

$\lambda^*_{max}/\lambda^*_{min} = -$

PROBLEM:	110
CLASSIFICATION:	GBR-P1-1
SOURCE:	Himmelblau [29], Paviani [48]
NUMBER OF VARIABLES:	$n = 10$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 0$, $b = 20$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{i=1}^{10} [(\ln(x_i - 2))^2 + (\ln(10 - x_i))^2 - (\prod_{i=1}^{10} x_i)^{.2}$$

CONSTRAINTS:

$$2.001 \leq x_i \leq 9.999 \quad , \quad i=1,\ldots,10$$

START:	x_o	=	(9,...,9) (feasible)
	$f(x_o)$	=	-43.134337
SOLUTION:	x^*	=	(9.35025655,...,9.35025655)
	$f(x^*)$	=	-45.77846971
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	0
	$I(x^*)$	=	-
	u^*_{max}/u^*_{min}	=	-
	$\lambda^*_{max}/\lambda^*_{min}$	=	6.92/6.52 = 1.06

PROBLEM:	111
CLASSIFICATION:	GGR-P1-4
SOURCE:	Bracken, McCormick [13], Himmelblau [29], White [58]
NUMBER OF VARIABLES:	$n = 10$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3$, $b = 20$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{j=1}^{10} \exp(x_j)(c_j + x_j - \ln(\sum_{k=1}^{10} \exp(x_k)))$$

c_j : cf. Appendix A

CONSTRAINTS:

$$\exp(x_1) + 2\exp(x_2) + 2\exp(x_3) + \exp(x_6) + \exp(x_{10}) - 2 = 0$$

$$\exp(x_4) + 2\exp(x_5) + \exp(x_6) + \exp(x_7) - 1 = 0$$

$$\exp(x_3) + \exp(x_7) + \exp(x_8) + 2\exp(x_9) + \exp(x_{10}) - 1 = 0$$

$$-100 \le x_i \le 100 \quad , \quad i=1,\ldots,10$$

START: $x_o = (-2.3 , \ldots ,-2.3)$ (not feasible)

$f(x_o) = -21.015$

SOLUTION: $x^* = (-3.201212,-1.912060,-.2444413,-6.537489,$
$-.7231524,-7.267738,-3.596711,-4.017769,$
$-3.287462,-2.335582)$

$f(x^*) = -47.76109026$

$r(x^*) = .34E-9$ $e(x^*) = .14E-3$

$\mu = 0$ $I(x^*) = -$

$u^*_{max}/u^*_{min} = 15.22/9.785 = 1.56$

$\lambda^*_{max}/\lambda^*_{min} = .11/.70E-3 = 160.1$

PROBLEM:	112 (chemical equilibrium)
CLASSIFICATION:	GLR-P1-3
SOURCE:	Bracken, McCormick [13], Himmelblau [29], White [58]
NUMBER OF VARIABLES:	$n = 10$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 3(3)$, $b = 10$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{j=1}^{10} x_j\left(c_j + \ln \frac{x_j}{x_1 + \ldots + x_{10}}\right)$$

c_j : cf. Appendix A

CONSTRAINTS:

$$x_1 + 2x_2 + 2x_3 + x_6 + x_{10} - 2 = 0$$

$$x_4 + 2x_5 + x_6 + x_7 - 1 = 0$$

$$x_3 + x_7 + x_8 + 2x_9 + x_{10} = 0$$

$$1.E-6 \le x_i \quad , \quad i=1,\ldots,10$$

START: $x_o = (.1 , \ldots , .1)$ (not feasible)

$f(x_o) = -20.961$

SOLUTION: $x^* = (.01773548,.08200180,.8825646,.7233256E-3,$
$.4907851,.4335469E-3,.01727298,$
$.007765639,.01984929,.05269826)$

$f(x^*) = -47.707579$

$r(x^*) = .23E-7$ $e(x^*) = .43E-6$

$\mu = 2$ $I(x^*) = (4 , 6)$

$u^*_{max}/u^*_{min} = 15.02/.262E-3 = .57E5$

$\lambda^*_{max}/\lambda^*_{min} = 191/8.98 = 21.3$

PROBLEM:	113 (Wong No.2)
CLASSIFICATION:	QQR-P1-7
SOURCE:	Asaadi [1], Charalambous [18], Wong [59]
NUMBER OF VARIABLES:	$n = 10$
NUMBER OF CONSTRAINTS:	$m_1 = 8(3)$, $m-m_1 = 0$, $b = 0$

OBJECTIVE FUNCTION:

$$f(x) = x_1^2 + x_2^2 + x_1x_2 - 14x_1 - 16x_2 + (x_3 - 10)^2 + 4(x_4 - 5)^2 + (x_5 - 3)^2 + 2(x_6 - 1)^2 + 5x_7^2 + 7(x_8 - 11)^2 + 2(x_9 - 10)^2 + (x_{10} - 7)^2 + 45$$

CONSTRAINTS:

$$105 - 4x_1 - 5x_2 + 3x_7 - 9x_8 \geq 0$$
$$-10x_1 + 8x_2 + 17x_7 - 2x_8 \geq 0$$
$$8x_1 - 2x_2 - 5x_9 + 2x_{10} + 12 \geq 0$$
$$-3(x_1 - 2)^2 - 4(x_2 - 3)^2 - 2x_3^2 + 7x_4 + 120 \geq 0$$
$$-5x_1^2 - 8x_2 - (x_3 - 6)^2 + 2x_4 + 40 \geq 0$$
$$-.5(x_1 - 8)^2 - 2(x_2 - 4)^2 - 3x_5^2 + x_6 + 30 \geq 0$$
$$-x_1^2 - 2(x_2 - 2)^2 + 2x_1x_2 - 14x_5 + 6x_6 \geq 0$$
$$3x_1 - 6x_2 - 12(x_9 - 8)^2 + 7x_{10} \geq 0$$

START: $x_o = (2,3,5,5,1,2,7,3,6,10)$ (feasible)

$f(x_o) = 753$

SOLUTION: $x^* = (2.171996, 2.363683, 8.773926, 5.095984, .9906548, 1.430574, 1.321644, 9.828726, 8.280092, 8.375927)$

$f(x^*) = 24.3062091$

$r(x^*) = .12E-8$ $e(x^*) = .46E-9$

$\mu = 6$ $I(x^*) = (1,2,3,4,5,7)$

$u^*_{max}/u^*_{min} = 1.717/.02055 = 83.5$

$\lambda^*_{max}/\lambda^*_{min} = 7.79/2.24 = 3.48$

PROBLEM:	114 (alkylation process)
CLASSIFICATION:	QGR-P1-6
SOURCE:	Bracken, McCormick [13]
NUMBER OF VARIABLES:	$n = 10$
NUMBER OF CONSTRAINTS:	$m_1 = 8(4)$, $m-m_1 = 3(1)$, $b = 20$

OBJECTIVE FUNCTION:

$$f(x) = 5.04x_1 + .035x_2 + 10x_3 + 3.36x_5 - .063x_4x_7$$

CONSTRAINTS:

$$g_1(x) = 35.82 - .222x_{10} - bx_9 \geq 0$$
$$g_2(x) = -133 + 3x_7 - ax_{10} \geq 0$$
$$g_3(x) = -g_1(x) + x_9(1/b - b) \geq 0$$
$$g_4(x) = -g_2(x) + (1/a - a)x_{10} \geq 0$$
$$g_5(x) = 1.12x_1 + .13167x_1x_8 - .00667x_1x_8^2 - ax_4 \geq 0$$
$$g_6(x) = 57.425 + 1.098x_8 - .038x_8^2 + .325x_6 - ax_7 \geq 0$$
$$g_7(x) = -g_5(x) + (1/a - a)x_4 \geq 0$$
$$g_8(x) = -g_6(x) + (1/a - a)x_7 \geq 0$$
$$g_9(x) = 1.22x_4 - x_1 - x_5 = 0$$
$$g_{10}(x) = 98000x_3/(x_4x_9 + 1000x_3) - x_6 = 0 \qquad a = .99$$
$$g_{11}(x) = (x_2 + x_5)/x_1 - x_8 = 0 \qquad b = .9$$

bounds: cf. Appendix A

START: x_o = (1745,12000,110,3048,1974,89.2,92.8,8,
$f(x_o)$ = -872.3872 \ 3.6,145) (not feasible)

SOLUTION: x^* = (1698.096,15818.73,54.10228,3031.226,
2000,90.11537,95,10.49336,1.561636,
153.53535)

$f(x^*)$ = -1768.80696

$r(x^*)$ = 0 $e(x^*)$ = .16E-5

u = 6 $I(x^*)$ = (2,3,5,6,23,25)

u^*_{max}/u^*_{min} = 311.8/.6778 = 460

$\lambda^*_{max}/\lambda^*_{min}$ = .14E-4/.14E-4 = 1

PROBLEM:	116 (3-stage membrane separation)
CLASSIFICATION:	LQR-P1-6
SOURCE:	Dembo [21,22]
NUMBER OF VARIABLES:	$n = 13$
NUMBER OF CONSTRAINTS:	$m_1 = 15(5)$, $m-m_1 = 0$, $b = 26$

OBJECTIVE FUNCTION:

$$f(x) = x_{11} + x_{12} + x_{13}$$

CONSTRAINTS:

$$x_3 - x_2 \geq 0 \qquad x_2 - x_1 \geq 0$$

$$1 - .002x_7 + .002x_8 \geq 0 \qquad 50 \leq f(x) \leq 250$$

$$x_{13} - 1.262626x_{10} + 1.231059x_3x_{10} \geq 0$$

$$x_5 - .03475x_2 - .975x_2x_5 + .00975x_2^2 \geq 0$$

$$x_6 - .03475x_3 - .975x_3x_6 + .00975x_3^2 \geq 0$$

$$x_5x_7 - x_1x_8 - x_4x_7 + x_4x_8 \geq 0$$

$$1 - .002(x_2x_9 + x_5x_8 - x_1x_8 - x_6x_9) - x_5 - x_6 \geq 0$$

$$x_2x_9 - x_3x_{10} - x_6x_9 - 500x_2 + 500x_6 + x_2x_{10} \geq 0$$

$$x_2 - .9 - .002(x_2x_{10} - x_3x_{10}) \geq 0$$

$$x_4 - .03475x_1 - .975x_1x_4 + .00975x_1^2 \geq 0$$

$$x_{11} - 1.262626x_8 + 1.231059x_1x_8 \geq 0$$

$$x_{12} - 1.262626x_9 + 1.231059x_2x_9 \geq 0$$ bounds: cf. Appendix A

START: x_o = (.5,.8,.9,.1,.14,.5,489,80,650,450,150,
$f(x_o)$ = 450 \ 150,150) (not feasible)

SOLUTION: x^* = (.8037703,.8999860,.9709724,.09999952,
.1908154,.4605717,574.0803,74.08043,
500.0162,.1,20.23413,77.34755,.00673039)

$f(x^*)$ = 97.588409

$r(x^*)$ = 0 $e(x^*)$ = 0

u = 14 $I(x^*)$ = (3,6,...,15,25,28,
u^*_{max}/u^*_{min} = 2088/.423E-3 = .49E7 \ 32)

$\lambda^*_{max}/\lambda^*_{min}$ = -

PROBLEM:	117 (Colville No.2, Shell Dual)
CLASSIFICATION:	PQR-P1-1
SOURCE:	Colville [20], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 15$
NUMBER OF CONSTRAINTS:	$m_1 = 5$, $m-m_1 = 0$, $b = 15$

OBJECTIVE FUNCTION:

$$f(x) = -\sum_{j=1}^{10} b_j x_j + \sum_{j=1}^{5} \sum_{k=1}^{5} c_{kj} x_{10+k} x_{10+j} + 2\sum_{j=1}^{5} d_j x_{10+j}^3$$

CONSTRAINTS:

$$2\sum_{k=1}^{5} c_{kj} x_{10+k} + 3d_j x_{10+j}^2 + e_j - \sum_{k=1}^{10} a_{kj} x_k \geq 0 \quad , \quad j=1,\ldots,5$$

$$0 \leq x_i \quad , \quad i=1,\ldots,15$$

$a_{ij}, b_j, c_{ij}, d_j, e_j$: cf. Appendix A

START:	x_o	=	.001(1,1,1,1,1,1,60000,1,1,1,1,1,1,1,1)
	$f(x_o)$	=	2400.1053 (feasible)
SOLUTION:	x^*	=	(0,0,5.174136,0,3.061093,11.83968,0,0, .1039071,0,.2999929,.3334709,.3999910, .4283145,.2239607)
	$f(x^*)$	=	32.348679
	$r(x^*)$	=	0 $e(x^*)$ = .35E-4
	u	=	11 $I(x^*)$ = (1,...,7,9,12,13, \ 15)
	u^*_{max}/u^*_{min}	=	56.75/.2240 = 253
	$\lambda^*_{max}/\lambda^*_{min}$	=	3.30/.10 = 32.3

PROBLEM:	118
CLASSIFICATION:	QLR-P1-2
SOURCE:	Bartholomew-Biggs [4]
NUMBER OF VARIABLES:	$n = 15$
NUMBER OF CONSTRAINTS:	$m_1 = 29(29)$, $m-m_1 = 0$, $b = 30$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{k=0}^{4} (2.3x_{3k+1} + .0001x_{3k+1}^2 + 1.7x_{3k+2} + .0001x_{3k+2}^2 + 2.2x_{3k+3} + .00015x_{3k+3}^2)$$

CONSTRAINTS:

$0 \le x_{3j+1} - x_{3j-2} + 7 \le 13$ $\quad$ $0 \le x_{3j+2} - x_{3j-1} + 7 \le 14$

$0 \le x_{3j+3} - x_{3j} + 7 \le 13$ $\quad$ $j=1,\ldots,4$

$x_1 + x_2 + x_3 - 60 \ge 0$ $\quad$ $x_4 + x_5 + x_6 - 50 \ge 0$

$x_7 + x_8 + x_9 - 70 \ge 0$ $\quad$ $x_{10} + x_{11} + x_{12} - 85 \ge 0$

$x_{13} + x_{14} + x_{15} - 100 \ge 0$

$8 \le x_1 \le 21$ $\quad$ $43 \le x_2 \le 57$ $\quad$ $3 \le x_3 \le 16$

$0 \le x_{3k+1} \le 90$ $\quad$ $0 \le x_{3k+2} \le 120$ $\quad$ $0 \le x_{3k+3} \le 60$

$k=1,\ldots,4$

START:	x_o	=	(20,55,15,20,60,20,20,60,20,20,60, 20,20,60,20) (feasible)
	$f(x_o)$	=	664.82045000
SOLUTION:	x^*	=	(8,49,3,1,56,0,1,63,6,3,70,12, 5,77,18)
	$f(x^*)$	=	664.8204500
	$r(x^*)$	=	0
	$e(x^*)$	=	0
	μ	=	15
	$I(x^*)$	=	(1,17,18,19,20,22,23,24,25,27,28,29,30, 32,35)
	u^*_{max}/u^*_{min}	=	2.941/.04860 = 60.5
	$\lambda^*_{max}/\lambda^*_{min}$	=	-

PROBLEM:	119 (Colville No.7)
CLASSIFICATION:	PLR-P1-2
SOURCE:	Colville [20], Himmelblau [29]
NUMBER OF VARIABLES:	$n = 16$
NUMBER OF CONSTRAINTS:	$m_1 = 0$, $m-m_1 = 8(8)$, $b = 32$

OBJECTIVE FUNCTION:

$$f(x) = \sum_{i=1}^{16} \sum_{j=1}^{16} a_{ij}(x_i^2 + x_i + 1)(x_j^2 + x_j + 1)$$

CONSTRAINTS:

$$\sum_{j=1}^{16} b_{ij}x_j - c_i = 0 \quad , \quad i=1,\ldots,8$$

$$0 \le x_i \le 5 \quad , \quad i=1,\ldots,16$$

a_{ij}, b_{ij}, c_i : cf. Appendix A

START: $x_o = (10,\ldots,10)$ (not feasible)

$f(x_o) = 566766$

SOLUTION: $x^* = (.03984735, .7919832, .2028703, .8443579, 1.126991, .9347387, 1.681962, .1553009, 1.567870, 0, 0, 0, .6602041, 0, .6742559, 0)$

$f(x^*) = 244.899698$

$r(x^*) = .26E-9$ $e(x^*) = .36E-8$

$\mu = 5$ $I(x^*) = (10,11,12,14,16)$

$u^*_{max}/u^*_{min} = 95.99/4.201 = 22.9$

$\lambda^*_{max}/\lambda^*_{min} = 39.2/25.1 = 1.56$